arXiv:2405.17784v2 [cs.LG] 3 Jun 2024

Adaptive Horizon Actor-Critic for Policy Learning in Contact-Rich Differentiable Simulation

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Abstract

Model-Free Reinforcement Learning (MFRL), leveraging the policy gradient theorem, has demonstrated considerable success in continuous control tasks. However, these approaches are plagued by high gradient variance due to zeroth-order gradient estimation, resulting in suboptimal policies. Conversely, First-Order Model-Based Reinforcement Learning (FO-MBRL) methods employing differentiable simulation provide gradients with reduced variance but are susceptible to sampling error in scenarios involving stiff dynamics, such as physical contact. This paper investigates the source of this error and introduces Adaptive Horizon Actor-Critic (AHAC), an FO-MBRL algorithm that reduces gradient error by adapting the model-based horizon to avoid stiff dynamics. Empirical findings reveal that AHAC outperforms MFRL baselines, attaining 40% more reward across a set of locomotion tasks and efficiently scaling to high-dimensional control environments with improved wall-clock-time efficiency. adaptive-horizon-actor-critic.github.io

1. Introduction

The Policy Gradients Theorem (Sutton et al., 1999) has enabled the development of Model-Free Reinforcement Learning (MFRL) approaches for solving continuous motor control tasks. Although these methods have achieved impressive results (Hwangbo et al., 2017; Akkaya et al., 2019; Hwangbo et al., 2019), they are hampered by high gradient variance leading to unstable learning and suboptimal policies (Mohamed et al., 2020), as well as subpar sample efficiency (Amos et al., 2021). The latter can be circumvented via the use of efficient vectorized physics simulators. These simulators, when integrated with efficient MFRL methods, facilitate rapid training such as learning quadruped walking in minutes (Rudin et al., 2022). However, the effectiveness of MFRL in addressing motor control challenges, even with extensive data, remains questionable.

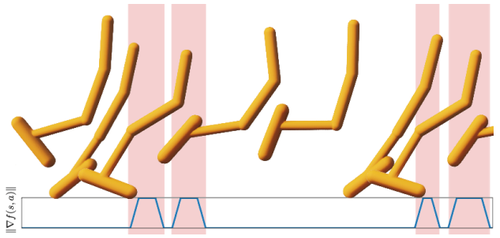


Figure 1. **Overview.** We find that First Order Model-Based RL (FO-MBRL) methods suffer from erroneous gradients arising from stiff dynamics ($\|\nabla f(s,a)\| \gg 0$). Our proposed method, *AHAC*, truncates model-based trajectories at the point of contact, avoiding both the gradient sample error and learning instability exhibited by previous methods using differentiable simulation.

An alternative, Model-Based Reinforcement Learning (MBRL), focuses on learning environmental dynamics to enhance sample efficiency and facilitate novel methods of policy optimization. Recent MBRL research has introduced innovative dynamics models and policy learning techniques, but often without extensive independent evaluation of each component (Hafner et al., 2019b; 2023; Hansen et al., 2023).

When a dynamics model is available, one could employ first-order methods for policy learning, deemed theoretically more efficient (Mohamed et al., 2020; Berahas et al., 2022). This approach has been investigated in model-based control, where dynamics models guide trajectory planning (Kabzan et al., 2019; Kaufmann et al., 2020). However, using first-order methods to learn feedback policies within typical MBRL frameworks is less explored. This paper aims to evaluate First-Order MBRL (FO-MBRL), concentrating on policy learning and utilizing differentiable simulation for dynamics modeling.

Where model-based control literature often designs bespoke models for each problem, differentiable simulation aims to create a physics engine that is fully differentiable (Hu

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Proceedings of the 41st International Conference on Machine Learning, Vienna, Austria. PMLR 235, 2024. Copyright 2024 by the author(s).

arXiv:2405.17784v2 [cs.LG] 3 Jun 2024

用于丰富接触可微分仿真中策略学习的自适应视界演员 - 评论家

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摘要

无模型强化学习（MFRL）利用策略梯度定理，在连续控制任务中取得了显著成功。然而，由于采用零阶梯度估计，这些方法深受高梯度方差之苦，导致策略次优。相反，采用可微分仿真的第一阶基于模型的强化学习（FO-MBRL）方法能够提供方差更低的梯度，但在涉及刚性动力学（如物理接触）的场景中容易受到采样误差的影响。本文研究了该误差的来源，并提出了自适应视界演员 - 评论家（AHAC），这是一种 FO-MBRL 算法，通过自适应调整基于模型的视界以避开刚性动力学，从而减少梯度误差。实证结果表明，AHAC 的表现优于免模型强化学习基线，在一系列运动任务中奖励提升了 40%，并能高效扩展至高维控制环境，同时提高了实际运行时间效率。

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1. 引言

策略梯度定理（Sutton 等，1999）推动了无模型强化学习（MFRL）方法的发展，用于解决连续运动控制任务。尽管这些方法已取得令人瞩目的成果（Hwangbo 等，2017；Akkaya 等，2019；Hwangbo 等，2019），但它们受限于高梯度方差，导致学习不稳定且策略次优（Mohamed etal.，2020），同时样本效率也欠佳（Amos 等，2021）。后者可通过使用高效的向量化物理仿真器来规避。这些仿真器在与高效的 MFRL 方法结合时，能够加速训练，例如在数分钟内学会四足行走（Rudin 等，2022）。然而，即使拥有大量数据，MFRL 在解决运动控制挑战方面的有效性仍存疑。

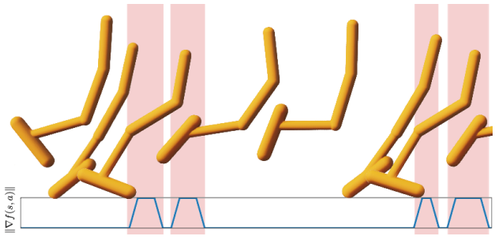


图 *1.***概述**。我们发现，一阶基于模型的强化学习（FO-MBRL）方法会因刚性动力学而产生错误的梯度（$\|\nabla f(s,a)\| \gg 0$）。我们提出的方法*AHAC*会在接触点截断基于模型的轨迹，从而避免了先前使用可微分仿真的方法所表现出的梯度样本误差和学习不稳定性。

另一种方法是基于模型的强化学习（MBRL），其侧重于学习环境动力学以提高样本效率并促进新颖的策略优化方法。近期的 MBRL 研究引入了创新的动力学模型和策略学习技术，但往往缺乏对每个组件的广泛独立评估（Hafner 等人，2019b；2023；Hansen 等人，2023）。

当具备动力学模型时，可采用一阶方法进行策略学习，该方法被认为在理论上更高效（Mohamed 等，2020；Berahas 等，2022）。这一方法已在基于模型的控制中得到研究，其中动力学模型用于指导轨迹规划（Kabzan 等，2019；Kaufmann 等，2020）。然而，在典型的基于模型的强化学习框架中，利用一阶方法学习反馈策略的研究尚不充分。本文旨在评估一阶基于模型的强化学习（FO-MBRL），重点关注策略学习，并利用可微分仿真进行动力学建模。

基于模型的控制文献通常为每个问题设计专用模型，而可微分仿真旨在创建一个完全可微分的物理引擎（Hu

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et al., 2019a; Freeman et al., 2021; Heiden et al., 2021; Xu et al., 2021). Thus, applying it to a different problem is similar to using a different definition of the environment in the simulation setup (e.g., joints and links) and leaving the physics to be calculated by the engine. Short Horizon Actor-Critic (SHAC) (Xu et al., 2022) is an FO-MBRL approach leveraging differentiable simulation and the popular actor-critic paradigm (Konda & Tsitsiklis, 1999). The actor is trained in a first-order fashion, while the critic is trained model-free. This allows SHAC to learn through the highly non-convex landscape by using the critic as a smooth surrogate of the cumulative reward. While SHAC demonstrates impressive sample efficiency, it also faces challenges such as brittleness, learning instability, and dependency on extensive hyper-parameter tuning.

This study addresses these issues, shifting the focus from sample efficiency to the asymptotic performance of FO-MBRL methods in massively parallel differentiable simulation. Our analysis indicates that first-order methods suffer from significant sampling error in gradient estimation, primarily due to high dynamical gradients from stiff contact approximation ($\|\nabla f(\boldsymbol{s}, \boldsymbol{a})\| \gg 0$) (Suh et al., 2022; Lee et al., 2023), leading to inefficiency and suboptimal policies. To address this, we propose Adaptive Horizon Actor-Critic (AHAC), a FO-MBRL algorithm that adjusts its trajectory rollout horizon to circumvent stiff dynamics (Figure 1). Experimentally, our method shows superior asymptotic performance over MFRL baselines in complex locomotion tasks, achieving up to 64% higher reward even when baselines are given 10^6 times more training data. Further, AHAC's efficient use of first-order gradients enables scaling to high-dimensional motor control tasks with 152 action dimensions.

2. Preliminaries

This study focuses on discrete-time and finite-horizon reinforcement learning scenarios characterized by system states $\boldsymbol{s} \in \mathbb{R}^n$, actions $\boldsymbol{a} \in \mathbb{R}^m$, and deterministic dynamics described by the function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$. Actions at each timestep t are sampled from a tanh-transformed stochastic policy $\boldsymbol{a}_t \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_t)$, parameterized by $\boldsymbol{\theta} \in \mathbb{R}^d$, and yield rewards from $r : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$. The H-step return is defined as:

$$R_H(\boldsymbol{s}_1, \boldsymbol{\theta}) = \sum_{h=1}^{H} r(\boldsymbol{s}_h, \boldsymbol{a}_h)$$
$$s.t. \quad \boldsymbol{s}_{h+1} = f(\boldsymbol{s}_h, \boldsymbol{a}_h) \quad \boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)$$

The policy's objective is to maximize the cumulative reward:

$$\max_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \max_{\boldsymbol{\theta}} \mathbb{E}_{\substack{\boldsymbol{s}_1 \sim \rho \\ \boldsymbol{a}_h \sim \pi(\cdot|\boldsymbol{s}_h)}} [R_H(\boldsymbol{s}_1)] \tag{1}$$

where ρ is the initial state distribution. Without loss of generality, we simplify our derivations:

Assumption 2.1. *ρ is a dirac-delta distribution.*

Similar to prior work Duchi et al. (2012); Berahas et al. (2022); Suh et al. (2022), we are trying to exploit the smoothing properties of stochastic optimization on the landscape of our optimization objective. Following recent successful deep-learning approaches to MFRL (Schulman et al., 2017; Haarnoja et al., 2018), we assume that our policy is stochastic, parameterized by $\boldsymbol{\theta}$ and expressed as $\pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s})$.

To address the main optimization problem in Equation 1, we consider stochastic gradient estimates of $J(\boldsymbol{\theta})$ using zero-order and first-order methods. To guarantee the existence of $\nabla J(\boldsymbol{\theta})$, we need to make certain assumptions:

Definition 2.2. *A function $g : \mathbb{R}^d \to \mathbb{R}^d$ has polynomial growth if there exists constants a, b such that $\forall \boldsymbol{z} \in \mathbb{R}^d$, $||g(\boldsymbol{z})|| \leq a(1 + ||\boldsymbol{z}||^b)$.*

Assumption 2.3. *To ensure gradients are well defined, we assume that the policy $\pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s})$ is continuously differentiable $\forall \boldsymbol{s} \in \mathbb{R}^n, \forall \boldsymbol{\theta} \in \mathbb{R}^d$. Furthermore, the system dynamics f and reward r have polynomial growth.*

2.1. Zeroth-Order Batch Gradient (ZOBG) estimates

These weak assumptions are sufficient to make $J(\boldsymbol{\theta})$ differentiable in expectation by taking samples of the function value in a zeroth-order fashion (Williams, 1992). This gives estimates of $\nabla J(\boldsymbol{\theta})$ via the stochasticity introduced by π, as first shown in (Williams, 1992), and commonly referred to as as the *Policy Gradient Theorem* (Sutton et al., 1999).

Definition 2.4. *Given a sample of the H-step return $R_H(\boldsymbol{s}_1) = \sum_{h=1}^{H} r(\boldsymbol{s}_h, \boldsymbol{a}_h)$ following the policy π, we can estimate zero-order policy gradients via:*

$$\nabla^{[0]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \mathbb{E}_{\boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)} \left[R_H(\boldsymbol{s}_1) \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \right] \tag{2}$$

Lemma 2.5. *Under Assumptions 2.1 and 2.3, the ZOBG is an unbiased estimator of the stochastic objective $\mathbb{E}\left[\bar{\nabla}^{[0]} J(\boldsymbol{\theta})\right] = \nabla J(\boldsymbol{\theta})$ where $\bar{\nabla}^{[0]} J(\boldsymbol{\theta})$ is the sample mean of N Monte Carlo estimates of Eq. 2.*

These zero-order policy gradients are known to have high variance (Mohamed et al., 2020), and one way to reduce their variance is by subtracting a baseline from the function estimates. Similar to (Suh et al., 2022), we subtract the return given by the noise-free policy rollout where $\left(R_H(\boldsymbol{s}_1) - R^*_H(\boldsymbol{s}_1)\right)$ is used instead of $R_H(\boldsymbol{s}_1)$ in Eq. 2.

2.2. First-Order Batch Gradient (FOBG) estimates

Given access to a differentiable simulator, first-order gradients induced by the policy π can be computed via:

$$\nabla^{[1]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \mathbb{E}_{\boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)} [\nabla_{\boldsymbol{\theta}} R_H(\boldsymbol{s}_1)] \tag{3}$$

等，2019a；Freeman 等，2021；Heiden 等，2021；Xu 等，2021）。因此，将其应用于不同问题时，类似于在仿真设置中使用不同的环境定义（例如关节和连杆），而由引擎负责计算物理学过程。短视界演员 - 评论家（SHAC）（Xu 等，2022）是一种 FO-MBRL 方法，利用可微分仿真和流行的演员 - 评论家范式（Konda 与 Tsitsiklis，1999）。其演员以一阶方式进行训练，而评论家则以无模型方式训练。这使得 SHAC 能够通过将评论家作为累积奖励的平滑代理，在高非凸景观中进行学习。尽管 SHAC 展现出令人印象深刻的样本效率，但也面临脆弱性、学习不稳定以及依赖大量超参数调优等挑战。

本研究针对这些问题，将焦点从样本效率转向 FO-MBRL 方法在大规模并行可微分仿真中的渐近性能。我们的分析表明，一阶方法在梯度估计中遭受显著的采样误差，这主要是由于刚性接触近似产生的高动态梯度所致（$\|\nabla f(\boldsymbol{s}, \boldsymbol{a})\| \gg 0$）(Suh 等，2022；Lee 等人，2023)，导致效率低下和策略次优。为此，我们提出了自适应时域演员 - 评论家算法（AHAC），这是一种 FO-MBRL 算法，通过调整其轨迹展开视界来规避刚性动力学（图1）。实验表明，我们的方法在复杂运动任务中展现出优于 MFRL 基线的渐近性能，即使基线获得更多 10^6 倍的训练数据，其奖励仍高出高达 64%。此外，AHAC 对一阶梯度的高效利用使其能够扩展至具有152个动作维度的高维运动控制任务。

2. 预备知识

本研究聚焦于离散时间和有限 Horizon 的强化学习场景，其特征包括系统状态$\boldsymbol{s} \in \mathbb{R}^n$、动作 $\boldsymbol{a} \in \mathbb{R}^m$，以及由函数 $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$描述的确定性动力学。每个时间步 t 的动作均采样自一个由 $\boldsymbol{\theta} \in \mathbb{R}^d$参数化的 tanh 变换的随机策略 $\boldsymbol{a}_t \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_t)$，并产生来自 $r : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$的奖励。H 步回报定义为：

$$R_H(\boldsymbol{s}_1, \boldsymbol{\theta}) = \sum_{h=1}^{H} r(\boldsymbol{s}_h, \boldsymbol{a}_h)$$
$$s.t. \quad \boldsymbol{s}_{h+1} = f(\boldsymbol{s}_h, \boldsymbol{a}_h) \quad \boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)$$

策略的目标是最大化累积奖励：

$$\max_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \max_{\boldsymbol{\theta}} \mathbb{E}_{\substack{\boldsymbol{s}_1 \sim \rho \\ \boldsymbol{a}_h \sim \pi(\cdot|\boldsymbol{s}_h)}} [R_H(\boldsymbol{s}_1)] \tag{1}$$

其中 ρ 是初始状态分布。不失一般性，我们简化推导过程：

假设 2.1。 ρ是一个狄拉克 - 德尔塔分布。

与先前的工作类似Duchi 等(2012)；Berahas 等(2022)；Suh 等(2022)，我们试图利用随机优化在优化目标景观上的平滑性质。继近期无模型强化学习（MFRL）的深度学习方法取得成功之后（Schulman 等，2017；Haarnoja 等，2018），我们假设我们的策略是随机的，由 参数化，并表示为 $(\cdot|\boldsymbol{s})$。

为解决方程1中的主要优化问题，我们考虑使用零阶和一阶方法对 $J(\boldsymbol{\theta})$ 进行随机梯度估计。为保证$\nabla J(\boldsymbol{\theta})$的存在性，我们需要做出某些假设：

定义 2.2。如果存在常数 a和 b 使得 $\forall \boldsymbol{z} \in \mathbb{R}^d$、$||g(\boldsymbol{z})|| \leq a(1 + ||\boldsymbol{z}||^b)$成立，则称函数 $g : \mathbb{R}^d \to \mathbb{R}^d$ 具有多项式增长。

假设 2.3。为确保梯度定义良好，我们假设策略 $\pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s})$ 是连续可微的$\forall \boldsymbol{s} \in \mathbb{R}^n, \forall \boldsymbol{\theta} \in \mathbb{R}^d$。此外，系统动力学 f和奖励 r 具有多项式增长。

2.1. 零阶批量梯度 (ZOBG) 估计

这些较弱的假设足以使 $J(\boldsymbol{\theta})$ 通过以零阶方式对函数值进行采样而在期望上可微（Williams，1992）。这通过由 π引入的随机性给出了 $\nabla J(\boldsymbol{\theta})$ 的估计，正如（Williams，1992）中首次展示的那样，通常被称为策略梯度定理（Sutton 等，1999）。

定义 2.4。给定遵循策略 π 的 H 步回报 $R_H(\boldsymbol{s}_1) = \sum_{h=1}^{H} r(\boldsymbol{s}_h, \boldsymbol{a}_h)$ 的一个样本，我们可以通过以下方式估计零阶策略梯度：

$$\nabla^{[0]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \mathbb{E}_{\boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)} \left[R_H(\boldsymbol{s}_1) \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \right] \tag{2}$$

引理 2.5。在假设2.1和2.3下，*ZOBG* 是随机目标的一个无偏估计量$\mathbb{E}\left[\bar{\nabla}^{[0]} J(\boldsymbol{\theta})\right] = \nabla J(\boldsymbol{\theta})$，其中 $\bar{\nabla}^{[0]} J(\boldsymbol{\theta})$ 是式 N 蒙特卡洛估计的样本均值2。

这些零阶策略梯度已知具有高方差（Mohamed 等，2020），而降低其方差的一种方法是从函数估计中减去一个基线。与（Suh 等，2022）类似，我们减去由无噪声策略 rollout 给出的回报，其中在式 $\left(R_H(\boldsymbol{s}_1) - R^*_H(\boldsymbol{s}_1)\right)$中使用 $R_H(\boldsymbol{s}_1)$代替 2。

2.2. 一阶批量梯度 (FOBG) 估计

在访问可微模拟器的情况下，由策略 π 诱导的一阶梯度可通过以下方式计算：

$$\nabla^{[1]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \mathbb{E}_{\boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)} [\nabla_{\boldsymbol{\theta}} R_H(\boldsymbol{s}_1)] \tag{3}$$

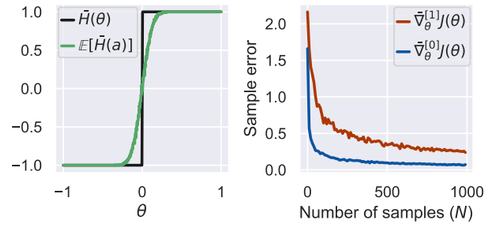


Figure 2. The left figure shows the **Soft Heaviside of Eq 4**. The right figure shows the gradient sample error. We observe that FOBG estimates with finite N exhibit a higher sample error.

However, for these gradients to be well-defined, we need to make further assumptions:

Assumption 2.6. *The dynamics $f(\boldsymbol{s}, \boldsymbol{a})$ and the reward $r(\boldsymbol{s}, \boldsymbol{a})$ are continuously differentiable $\forall \boldsymbol{s} \in \mathbb{R}^n, \forall \boldsymbol{a} \in \mathbb{R}^m$.*

Although these assumptions are necessary for the analysis of the next section, we relax them in our experiments section and consider contemporary benchmarks.

3. Policy learning through contact

Prior research has established that first-order gradients are statistically unbiased (Schulman et al., 2015). However, the sample error under finite N is heavily dependent on the function they are trying to approximate, referred to as "empirical bias" (Suh et al., 2022; Lee et al., 2023). This paper explores this sampling error using the soft Heaviside function, an approximation of the Coulomb friction model, which is pivotal in discontinuous function analysis within physics simulations:

$$\bar{H}(x) = \begin{cases} 1 & x > \nu/2 \\ 2x/\nu & |x| \leq \nu/2 \\ -1 & x < -\nu/2 \end{cases} \tag{4}$$

where $a \sim \pi_\theta(\cdot) = \theta + w$ and $w \sim \mathcal{N}(0, \sigma^2)$. As shown in Appendix A, $\mathbb{E}_\pi\big[\bar{H}(a)\big]$ is a sum of error functions whose derivative $\nabla_\theta \mathbb{E}_\pi\big[\bar{H}(a)\big] \neq 0$ at $\theta = 0$. However, using FOBG, we obtain $\nabla_\theta \bar{H}(a) = 0$ in samples where $|a| > \nu/2$, which occurs with probability at least $\nu/\sigma\sqrt{2\pi}$. Since in practice we are limited in sample size, this translates to sampling error that is inversely proportional to sample size, as shown in Figure 2. Notably, when $\nu \to 0$, we achieve a more accurate approximation of the underlying discontinuous function, but we also increase the likelihood of obtaining erroneous FOBG, thus amplifying error in stochastic scenarios. We use this particular example as the differentiable simulator used in our experiments is based on the Coulomb friction model.

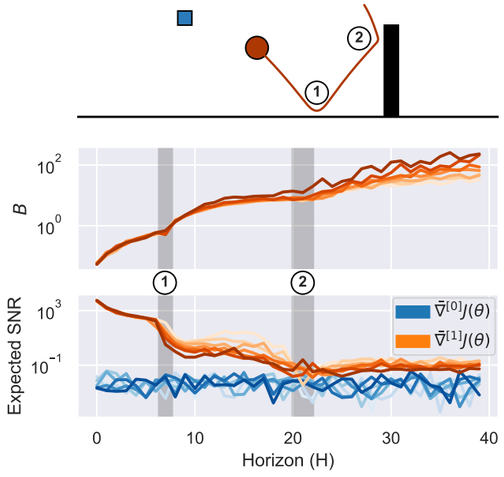


Figure 3. **Toy example where a ball is shot against a wall** trying to reach the target position in blue. The bottom two figures show gradient sample error and Expected SNR estimation with $N = 1024$ samples. Darker shades designate point of contact, which negatively impact FOBG error. Higher ESNR leads to more informative gradients.

Definition 3.1. *FOBGs exhibit sampling error relative to ZOBGs under finite samples, denoted as B:*

$$B = \left\| \bar{\nabla}^{[1]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) - \bar{\nabla}^{[0]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \right\|$$

We analyze B from the perspective of bias and variance to derive a practical upper bound:

Lemma 3.2. *For an H-step stochastic optimization problem under Assumptions 2.6, which also has Lipshitz-smooth policies $\|\nabla \pi_{\boldsymbol{\theta}}(\boldsymbol{a}|\boldsymbol{s})\| \leq B_\pi$, Lipshitz-smooth reward function in both arguments $\|\nabla r(\boldsymbol{s}, \boldsymbol{a})\| \leq B_r$ and Lipshitz-smooth dynamics in both arguments $\|\nabla f(\boldsymbol{s}, \boldsymbol{a})\| \leq B_f$ $\forall \boldsymbol{s} \in \mathbb{R}^n; \boldsymbol{a} \in \mathbb{R}^m; \boldsymbol{\theta} \in \mathbb{R}^d$, then ZOBGs remain consistently unbiased. However, FOBGs exhibit sample error bounded by:*

$$B \leq H B_r B_\pi \big(\frac{1}{2} + B_f^{H-1}\big) \tag{5}$$

The proof can be found in Appendix B

As $r(s, a)$ and π are often design decisions in problems, we can create them to satisfy the assumptions laid out above. However, bounding the dynamics $||\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)||$ is impossible due to the natural discontinuities of physics (Lee et al., 2023) leading to $B_f \gg B_r$ and $B_f \gg B_\pi$. This combined with the H terms lead the two conclusions from Lemma 3.2: (1) long-horizon rollout lead to increased FOBG sample error and (2) prolonged stiff contact has compound effects on FOBG sample error.

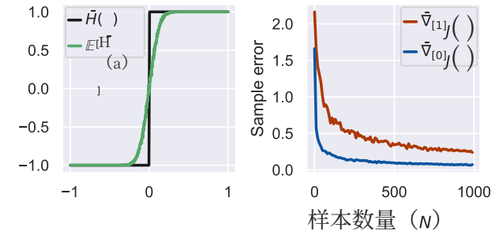


图 2。左图展示了公式**中的软赫维赛德函数4**。右图展示了梯度样本误差。我们观察到，使用有限 N 的 FOBG 估计表现出更高的样本误差。

然而，为了使这些梯度定义良好，我们需要做出进一步的假设：

假设 2.6。动力学 $f(\boldsymbol{s}, \boldsymbol{a})$ 和奖励 $r(\boldsymbol{s}, \boldsymbol{a})$ 是连续可微的 $\forall \boldsymbol{s} \in \mathbb{R}^n, \forall \boldsymbol{a} \in \mathbb{R}^m$。

尽管这些假设对于下一节的分析是必要的，但我们在实验部分放宽了这些假设，并考虑了当代基准测试。

3. 通过接触进行策略学习

先前的研究已确立，一阶梯度在统计上是无偏的（Schulman 等，2015）。然而，在有限 下的样本误差高度依赖于其试图逼近的函数，这被称为“经验偏差”（Suh 等，2022；Lee 等人，2023）。本文利用软赫维赛德函数（即库仑摩擦模型的近似）来探讨这种采样误差，该函数在物理仿真中的不连续函数分析中至关重要：

$$\bar{H}(x) = \begin{cases} 1 & x > \nu/2 \\ 2x/\nu & |x| \leq \nu/2 \\ -1 & x < -\nu/2 \end{cases} \tag{4}$$

其中 $a \sim \pi_\theta(\cdot) = \theta + w$ 且 $w \sim \mathcal{N}(0, \sigma^2)$。如附录 A 所示， $\mathbb{E}_\pi\big[\bar{H}(a)\big]$ 是误差函数的和，其导数 $\nabla_\theta \mathbb{E}_\pi\big[\bar{H}(a)\big] \neq 0$ 在 $\theta = 0$处。然而，¯使用一阶梯度估计时，我们在样本中得到 $\nabla_\theta H(a) = 0$ ，其中$|a| > \nu/2$，这种情况发生的概率至少为 $\nu/\sigma\sqrt{2\pi}$。由于在实践中我们的样本量有限，这转化为与样本量成反比的采样误差，如图 2所示。值得注意的是，当$\nu \to 0$时，我们能够更准确地近似底层的不连续函数，但也增加了获得错误一阶梯度估计的可能性，从而在随机场景中放大误差。我们使用这个特定例子，是因为我们实验中使用的可微模拟器基于库仑摩擦模型。

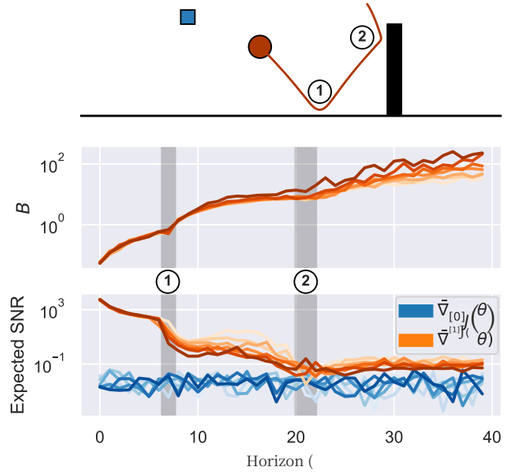


图 3.**一个玩具示例：一个小球被射向墙壁，试图到达蓝色的目标位置。底部两张图展示了使用 个样本时的梯度样本误差和期望信噪比估计。较深的阴影表示接触点，这会负面影响一阶梯度估计误差。更高的期望信噪比意味着梯度更具信息量。**

定义 3.1. 在有限样本下，一阶梯度估计相对于 *ZOBG* 表现出采样误差，记为 B：

$$B = \left\| \bar{\nabla}^{[1]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) - \bar{\nabla}^{[0]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \right\|$$

我们从偏差和方差的角度分析 B，以推导出一个实用的上界：

引理 3.2.对于假设2.6下的 H 步随机优化问题，若其还具有利普希茨平滑策略 $\|\nabla \pi_{\boldsymbol{\theta}}(\boldsymbol{a}|\boldsymbol{s})\| \leq B_\pi$、关于两个参数均为利普希茨平滑的奖励函数 $\|\nabla r(\boldsymbol{s}, \boldsymbol{a})\| \leq B_r$ 以及关于两个参数均为利普希茨平滑的动力学 $\|\nabla f(\boldsymbol{s}, \boldsymbol{a})\| \leq B_f$ $\forall \boldsymbol{s} \in \mathbb{R}^n; \boldsymbol{a} \in \mathbb{R}^m; \boldsymbol{\theta} \in \mathbb{R}^d$，则 *ZOBG* 始终保持无偏。然而，*FOBG* 的样本误差受限于：

$$B \leq H B_r B_\pi \big(\frac{1}{2} + B_f^{H-1}\big) \tag{5}$$

证明见附录*B*

由于 $r(s,\ a)$ 和 π 通常是问题中的设计决策，我们可以构建它们以满足上述假设。然而，由于物理学的自然不连续性（Lee 等人，2023）导致 $B_f \gg B_r$ 和 $B_f \gg B_\pi$，因此对动力学 $||\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)||$ 进行界定是不可能的。结合 H 项，这引出了引理 3.2的两个结论：(1) 长时程 rollout 会导致增加的 FOBG 采样误差；(2) 持续的刚性接触会对 FOBG 采样误差产生复合影响。

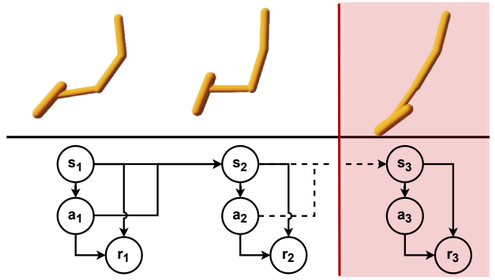


Figure 4. **Example $H = 3$ step trajectory** where s_3 is in contact at which point the trajectory is truncated. When optimizing this trajectory, we completely omit the stiff dynamics gradient $\nabla f(s_2, a_2)$ leading to stabler and less erroneous FOBGs.

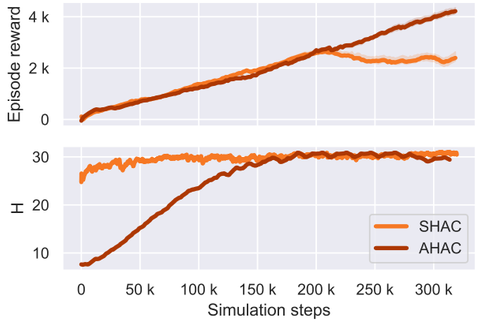


Figure 5. **Comparison between SHAC and AHAC-1 on the Hopper task with only a single environment**. The figure shows rewards and horizons achieved over 5 different random seeds, with the 50% IQM plotted. Note that both algorithms have some horizon oscillation due to the early termination mechanism of the simulator, as noted in Appendix F.

Empirical evaluation of Lemma 3.2. We designed a simple experimental setup involving a ball rebounding off a wall to reach a target, as illustrated in Figure 3. The initial position $s_1 = [x_1, y_1]$ and velocity of the ball are fixed. The objective is for the policy to learn the optimal initial orientation θ in order to reach a target position s_T at the end, defined as $R_H(s_1) = \|s_H - s_T\|_2^{-1}$. We use the additive Gaussian policy $a = \theta + w$, where $w \sim \mathcal{N}(0, \sigma^2)$. With this, zero-order gradients from Eq. 2 can be expressed as:

$$\nabla^{[0]}_{\boldsymbol{\theta}} J(\theta) \approx \frac{1}{N\sigma^2} \sum_{i=1}^{N} \left(R_H^{(i)}(s_1) - R_H^{*(i)}(s_1)\right) w^{(i)}$$

We collect $N = 1024$ samples of each gradient type for each timestep with $H = 40$. Figure 3 shows that the sample error remains low until the ball encounters contact, after which it starts growing, validating our proposed lemma. Additionally, the error also affects the gradient variance, where ZOBG follow $\mathrm{Var}\left[\nabla^{[0]} J(\theta)\right] \leq \sigma^{-2} H B_r^2 B_\pi^2$ (Suh et al., 2022). However, FOBG variance behaves similarly to Lemma 3.2, growing exponentially after contact. In Figure 3, instead of variance, we show Expected SNR (Eq. 6) as proposed by (Parmas et al., 2023), with higher values translating to more informative gradients. These results suggest that FOBGs exhibit sample error under contact dynamics, which is further worsened with long trajectories (Lee et al., 2023; Zhong et al., 2023).

$$\mathrm{ESNR}(\nabla J(\boldsymbol{\theta})) = \mathbb{E}\left[\frac{\sum \mathbb{E}[\nabla J(\boldsymbol{\theta})]^2}{\sum \mathrm{Var}[\nabla J(\boldsymbol{\theta})]}\right] \tag{6}$$

4. Adaptive Horizon Actor-Critic (AHAC)

4.1. Learning through contact in a single environment

With a clearer understanding of the influence of stiff contact, we aim to develop a First-Order MBRL approach for contact-rich continuous control tasks. Unlike the toy example of the previous section, standard RL multi-step decision processes allow for the avoidance of stiff dynamics gradients using *contact truncation*. Consider the example shown in Figure 4. Truncating the trajectory at the point of contact yields reward gradients without the gradient of stiff dynamics (striked out in red):

$$\nabla_{\boldsymbol{\theta}} r(s_3, a_3) = \nabla_{a_3} r(s_3, a_3) \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(a_3 | s_3) + \cancel{\nabla_{s_3} r(s_3, a_3) \nabla_{\boldsymbol{\theta}} f(s_2, a_2)}$$

We introduce an FO-MBRL algorithm with an actor-critic architecture, akin to SHAC (Xu et al., 2022). The critic, denoted as $V_\psi(s)$, is model-free and trained using TD(λ) (Sutton & Barto, 2018) over an H-step horizon:

$$R_h(s_t) := \sum_{n=t}^{t+h-1} \gamma^{n-t} r(s_n, a_n) + \gamma^{t+h} V_\psi(s_{t+h})$$

$$\hat{V}(s_t) := (1-\lambda)\left[\sum_{h=1}^{H-t-1} \lambda^{h-1} R_h(s_t)\right] + \lambda^{H-t-1} R_H(s_t)$$

The critic loss becomes $\mathcal{L}_V(\psi)$, while the actor is trained using FOBG as in Equation 3, with the addition of the critic value estimate:

$$\mathcal{L}_V(\psi) := \sum_{h=t}^{t+H} \left\| V_\psi(s_h) - \hat{V}(s_h) \right\|_2^2 \tag{7}$$

$$J(\boldsymbol{\theta}) := \sum_{h=t}^{t+H-1} \gamma^{h-t} r(s_h, a_h) + \gamma^H V_\psi(s_{t+T}) \tag{8}$$

Unlike fixed-horizon model-based rollouts in (Xu et al., 2022), our policy is rolled out until stiff contact is detected in simulation, leading to a dynamic horizon adjustment to prevent gradient explosion. However, not all contact results in high error; therefore, we truncate only on stiff

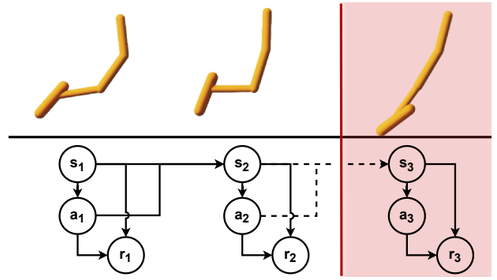


图 *4.*示例 $H = 3$ 步轨迹，其中 s_3 处于接触状态，此时轨迹被截断。在优化该轨迹时，我们完全省略了刚性动力学梯度 $\nabla f(s_2,\ a_2)$，从而获得更稳定且误差更小的一阶梯度估计。

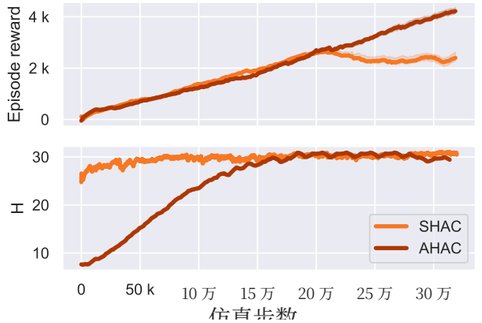


图 *5*。**SHAC 与 AHAC-1 在仅使用单一环境的 Hop-per 任务上的对比**。该图展示了在 5 个不同随机种子下获得的奖励和视界，绘制了 50% IQM。请注意，由于模拟器的早期终止机制（如附录F所述），两种算法均存在一定的视界振荡。

引理 3.2的实证评估。我们设计了一个简单的实验设置，涉及一个球从墙壁反弹以到达目标，如图 3所示。球的初始位置 $s_1 = [x_1, y_1]$ 和速度是固定的。目标是让策略学习最优的初始朝向 θ，以便最终到达定义为 s_T的目标位置 $R_H(s_1) = \|s_H - s_T\|_2^{-1}$。我们使用加性高斯策略 $a = \theta + w$，其中 $w \sim \mathcal{N}(0, \sigma^2)$。据此，公式 2中的零阶梯度可表示为：

$$\nabla^{[0]}_{\boldsymbol{\theta}} J(\theta) \approx \frac{1}{N\sigma^2} \sum_{i=1}^{N} \left(R_H^{(i)}(s_1) - R_H^{*(i)}(s_1)\right) w^{(i)}$$

我们收集 $N = 1024$ 每个时间步下每种梯度类型的样本，使用 $H = 40$。图3显示，样本误差在小球遇到接触之前保持较低，之后开始增长，验证了我们提出的引理。此外，该误差也会影响梯度方差，其中 Z OBG 遵循$\mathrm{Var}\left[\nabla^{[0]} J(\theta)\right] \leq \sigma^{-2} H B_r^2 B_\pi^2$ (Suh 等，2022)。然而，FOBG 的方差表现与引理3.2类似，在接触后呈指数增长。在图3中，我们未展示方差，而是展示了期望信噪比（公式6），正如（Parmas 等，2023）所提议的那样，较高的值意味着更具信息量的梯度。这些结果表明，FOBG 在接触动力学下会出现样本误差，且在长轨迹情况下会进一步恶化 (Lee 等，2023；Zhong 等，2023）。

$$\mathrm{ESNR}(\nabla J(\boldsymbol{\theta})) = \mathbb{E}\left[\frac{\sum \mathbb{E}[\nabla J(\boldsymbol{\theta})]^2}{\sum \mathrm{Var}[\nabla J(\boldsymbol{\theta})]}\right] \tag{6}$$

4. 自适应时域演员 - 评论家算法（AHAC）

4.1. 在单一环境中通过接触进行学习

在更清晰地理解刚性接触的影响之后，我们旨在开发一种用于丰富接触的连续控制任务的一阶基于模型的强化学习方法。与上一节的玩具示例不同，标准的强化学习多步决策过程能够通过接触截断来避免刚性动力学梯度。4。考虑图所示的示例。在接触点截断轨迹可得到不包含刚性动力学梯度的奖励梯度（红色划掉部分）：

$$\nabla_{\boldsymbol{\theta}} r(s_3, a_3) = \nabla_{a_3} r(s_3, a_3) \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(a_3 | s_3) + \cancel{\nabla_{s_3} r(s_3, a_3) \nabla_{\boldsymbol{\theta}} f(s_2, a_2)}$$

我们提出了一种具有演员 - 评论家架构的 FO-MBRL 算法，类似于 SHAC（Xu 等，2022）。评论家记为 $V_\psi(s)$，它是无模型的，并使用 TD(λ)（Sutton & Barto，2018）在 H步视界上进行训练：

$$R_h(s_t) := \sum_{n=t}^{t+h-1} \gamma^{n-t} r(s_n, a_n) + \gamma^{t+h} V_\psi(s_{t+h})$$

$$\hat{V}(s_t) := (1-\lambda)\left[\sum_{h=1}^{H-t-1} \lambda^{h-1} R_h(s_t)\right] + \lambda^{H-t-1} R_H(s_t)$$

评论家损失变为 $\mathcal{L}_V(\psi)$，而演员则按照公式3使用 FOBG 进行训练，并增加了评论家价值估计：

$$\mathcal{L}_V(\psi) := \sum_{h=t}^{t+H} \left\| V_\psi(s_h) - \hat{V}(s_h) \right\|_2^2 \tag{7}$$

$$J(\boldsymbol{\theta}) := \sum_{h=t}^{t+H-1} \gamma^{h-t} r(s_h, a_h) + \gamma^H V_\psi(s_{t+T}) \tag{8}$$

与（Xu 等，2022）中的固定视界基于模型的推出不同，我们的策略会一直推出，直到在仿真中检测到刚性接触，从而进行动态视界调整以防止梯度爆炸。然而，并非所有接触都会导致高误差；因此，我们仅在刚性

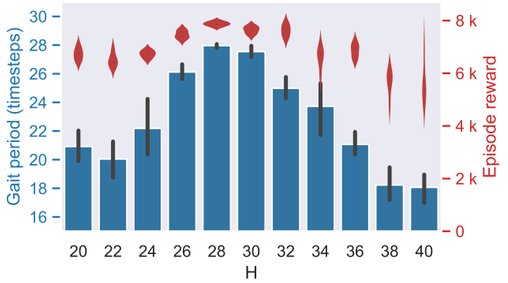


Figure 6. **An ablation of short horizons** H for the SHAC algorithm applied to Ant. Each run is trained until convergence for 5 seeds. The reward peaked and exhibited the least variance when the horizon length approximated the optimal gait period $H \approx 28$.

contact $\|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| > C$, where C is the contact stiffness threshold. We refer to this algorithm as Adaptive Horizon Actor-Critic 1 (AHAC-1) (see Appendix D).

AHAC-1's performance was tested in a toy locomotion environment. We re-implement the popular Hopper task, where a single-legged agent hops in one axis and is rewarded for high forward velocity (Figure 1). Compared to SHAC, which employs a fixed horizon of $H = 32$, AHAC-1 adjusts its horizon based on a contact stiffness threshold of $C = 500$. Results in Figure 5 indicate that AHAC-1 achieves a higher reward than SHAC. We believe that the more erroneous SHAC gradients steer it towards local minima, while our proposed approach manages to circumvent them and achieve a higher asymptotic reward. However, AHAC-1 is not applicable to parallel vectorized environments due to the challenge of asynchronous trajectory truncation, which leads to infinitely long compute graphs.

4.2. Scaling learning with synchronous parallelization

To address the issue of asynchronous truncation, we explored the short-horizon methodology of SHAC, incorporating graph truncation at stiff contacts. However, this method did not improve performance, likely due to gradient variances across differing trajectory lengths. Consequently, our research pivoted to examine the effect of horizon length H on policy optimality, especially in contact-based tasks like locomotion that demand specific gait patterns.

We conducted empirical tests using the SHAC algorithm on the Ant locomotion task, where the goal for a quadruped robot is to maximize forward velocity. By altering the horizon length H in SHAC, our findings in Figure 6 reveal a correlation between gait period and horizon length, with optimal performance at $H = 28$.

Two key insights emerged from this study: (1) each task possesses an inherent optimal model-based horizon length H, closely linked to the gait period; (2) the optimal horizon correlates with the highest reward and lowest variance, aligning with the findings of Lemma 3.2. These insights informed the development of a generalized, GPU-parallelized version of AHAC-1, termed *AHAC*. While retaining the same critic training methodology as outlined in Equation 7, AHAC introduces a novel constrained objective for the actor.

Algorithm 1 Adaptive Horizon Actor-Critic

Given: $\alpha_{\boldsymbol{\theta}}, \alpha_{\boldsymbol{\phi}}, \alpha_{\boldsymbol{\psi}}$: learning rates
Given: C: contact threshold
Initialize learnable parameters $\boldsymbol{\theta}, \boldsymbol{\psi}, H, \boldsymbol{\phi} = \mathbf{0}$
$t \leftarrow 0$
while episode not done **do**
  Initialize rollout buffer D
  for $h = 0, 1, .., H$ **do** ▷ rollout policy
    $\boldsymbol{a}_{t+h} \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_{t+h})$
    $r_{t+h} = r(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h})$
    $\boldsymbol{s}_{t+h+1} = f(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h})$
    $D \leftarrow D \cup \{(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h}, r_{t+h}, V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+h+1}))\}$
  end for
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \mathcal{L}_\pi(\boldsymbol{\theta}, \boldsymbol{\phi})$ ▷ train actor (Eq. 10)
  $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \alpha_{\boldsymbol{\phi}} \nabla_{\boldsymbol{\phi}} \mathcal{L}_\pi(\boldsymbol{\theta}, \boldsymbol{\phi})$
  $H \leftarrow H + \alpha_{\boldsymbol{\phi}} \sum_{h=0}^{H} \phi_h$
  while not converged **do** ▷ train critic (Eq. 7)
    sample $(\boldsymbol{s}, \hat{V}(\boldsymbol{s})) \sim D$
    $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} - \alpha_{\boldsymbol{\psi}} \nabla_{\boldsymbol{\psi}} \mathcal{L}_V(\boldsymbol{\psi})$
  end while
  $t \leftarrow t + H$
end while

$$J(\boldsymbol{\theta}) := \sum_{h=t}^{t+H-1} \gamma^{h-t} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \gamma^H V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+H}) \quad s.t. \quad \|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| \le C \quad \forall t \in \{0, .., H\} \tag{9}$$

The objective seeks to maximize the reward while ensuring that all contact stiffness remains below the C threshold. Using the Lagrangian formulation, we derive the dual problem:

$$\mathcal{L}_\pi(\boldsymbol{\theta}, \boldsymbol{\phi}) = \sum_{h=t}^{t+H-1} \gamma^{h-t} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \gamma^H V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+H}) + \boldsymbol{\phi}^T \left(C - \begin{bmatrix} \|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| \\ \vdots \\ \|\nabla f(\boldsymbol{s}_{t+H}, \boldsymbol{a}_{t+H})\| \end{bmatrix} \right) \tag{10}$$

By definition, $\phi_i = 0$ if the constraint is met and $\phi_i > 0$ otherwise. Thus, $\boldsymbol{\phi}$ is used to adapt the horizon, resulting in the full AHAC shown in Algorithm 1. Additionally, we introduce a double critic that is trained until convergence, defined as a small change in the last 5 critic training iterations, $\sum_{i=n-5}^{n} \mathcal{L}(\boldsymbol{\psi}) < 0.2$, where we take mini-batch samples from the rollout buffer $(\boldsymbol{s}, \hat{V}(\boldsymbol{s})) \sim D$.

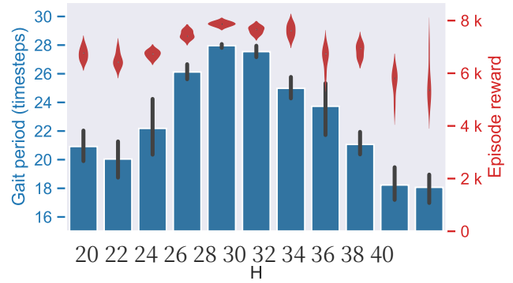


图 6。**对应用于蚂蚁的 SHAC 算法进行短 Horizon 的消融实验** H。每次运行均使用 5 个种子训练至收敛。当 Horizon 长度近似于最优步态周期时，奖励达到峰值且方差最小 $H \approx 28$。

接触 $\|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| > C$时截断，其中 C是接触刚度阈值。我们将此算法称为自适应视界演员 - 评论家 1（AHAC-1）（参见附录D）。

AHAC-1 的性能在玩具运动环境中进行了测试。我们重新实现了流行的跳跃者任务，其中单腿代理在一个轴上跳跃，并因高前进速度而获得奖励（图1）。与采用固定 Horizon 为 $H = 32$的 SHAC 相比，AHAC-1 根据 $C = 500$的接触刚度阈值调整其 Horizon。图5 的结果表明，AHAC-1 实现了比 SHAC 更高的奖励。我们认为，SHAC 中误差更大的梯度使其陷入局部极小值，而我们提出的方法能够规避这些问题并实现更高的渐近奖励。然而，由于异步轨迹截断的挑战会导致无限长的计算图，AHAC-1 不适用于并行向量化环境。

4.2. 通过同步并行化扩展学习

为解决异步截断问题，我们探索了 SHAC 的短 Horizon 方法，并在刚性接触处引入了图截断。然而，该方法并未提升性能，这可能是由于不同轨迹长度导致的梯度方差所致。因此，我们的研究转向考察 Horizon 长度 H 对策略最优性的影响，特别是在需要特定步态模式的基于接触的任务（如 locomotion）中。

我们在 Ant locomotion 任务上使用 SHAC 算法进行了实证测试，该任务的目标是最大化四足机器人的前进速度。通过改变 SHAC 中的 Horizon 长度 H，我们在 Figure6 中发现步态周期与 Horizon 长度之间存在相关性，且在 $H = 28$时性能最优。

本研究出现了两个关键见解：(1) 每个任务都具有一个固有的最优基于模型的 Horizon 长度H，其与步态周期密切相关；(2) 最优 Horizon 与最高奖励和最低方差相关，这与 Lemma 3.2的发现一致。这些见解促成了 AHAC-1 的通用、GPU 并行化版本的开发，称为 *AHAC*。在保留方程7中概述的相同 critic 训练方法的同时，AHAC 为 Actor 引入了一种新的约束目标。

Algorithm 1 Adaptive Horizon Actor-Critic

Given: $\alpha_{\boldsymbol{\theta}}, \alpha_{\boldsymbol{\phi}}, \alpha_{\boldsymbol{\psi}}$: learning rates
Given: C: contact threshold
Initialize learnable parameters $\boldsymbol{\theta}, \boldsymbol{\psi}, H, \boldsymbol{\phi} = \mathbf{0}$
$t \leftarrow 0$
while episode not done **do**
  Initialize rollout buffer D
  for $h = 0, 1, .., H$ **do** ▷ rollout policy
    $\boldsymbol{a}_{t+h} \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_{t+h})$
    $r_{t+h} = r(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h})$
    $\boldsymbol{s}_{t+h+1} = f(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h})$
    $D \leftarrow D \cup \{(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h}, r_{t+h}, V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+h+1}))\}$
  end for
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \mathcal{L}_\pi(\boldsymbol{\theta}, \boldsymbol{\phi})$ ▷ train actor (Eq. 10)
  $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \alpha_{\boldsymbol{\phi}} \nabla_{\boldsymbol{\phi}} \mathcal{L}_\pi(\boldsymbol{\theta}, \boldsymbol{\phi})$
  $H \leftarrow H + \alpha_{\boldsymbol{\phi}} \sum_{h=0}^{H} \phi_h$
  while not converged **do** ▷ train critic (Eq. 7)
    sample $(\boldsymbol{s}, \hat{V}(\boldsymbol{s})) \sim D$
    $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} - \alpha_{\boldsymbol{\psi}} \nabla_{\boldsymbol{\psi}} \mathcal{L}_V(\boldsymbol{\psi})$
  end while
  $t \leftarrow t + H$
end while

$$J(\boldsymbol{\theta}) := \sum_{h=t}^{t+H-1} \gamma^{h-t} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \gamma^H V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+H}) \quad s.t. \quad \|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| \le C \quad \forall t \in \{0, .., H\} \tag{9}$$

该目标旨在最大化奖励，同时确保所有接触刚度均低于 C 阈值。利用拉格朗日公式化，我们推导出 Dual 问题：

$$\mathcal{L}_\pi(\boldsymbol{\theta}, \boldsymbol{\phi}) = \sum_{h=t}^{t+H-1} \gamma^{h-t} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \gamma^H V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+H}) + \boldsymbol{\phi}^T \left(C - \begin{bmatrix} \|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| \\ \vdots \\ \|\nabla f(\boldsymbol{s}_{t+H}, \boldsymbol{a}_{t+H})\| \end{bmatrix} \right) \tag{10}$$

根据定义，$\phi_i = 0$ 若满足约束则取该值，否则取另一值。$\phi_i > 0$因此，$\boldsymbol{\phi}$ 被用于自适应调整 Horizon，从而形成算法1中所示的完整 AHAC。此外，我们引入了一个双重评判器，其训练直至收敛（定义为最近 5 次 critic 训练中变化很小），即 $\sum_{i=n-5}^{n} \mathcal{L}(\boldsymbol{\psi}) < 0.2$，其中我们从 rollout 缓冲区 $(\boldsymbol{s}, \hat{V}(\boldsymbol{s})) \sim D$中抽取小批量样本。



Figure 7. **Experimental Environments**. Locomotion tasks in increasing order of action space (m) dimensions (left to right): Hopper ($m = 3$), Ant ($m = 8$), Anymal ($m = 12$), Humanoid ($m = 21$) and SNU Humanoid ($m = 152$).

In practice, truncating based solely on the gradient of dynamics $\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)$ proved restrictive due to the variability of contact forces across different tasks and their evolution during the learning process. To address this, we introduced a normalization method for contact forces, utilizing modified acceleration per state dimension $\hat{\boldsymbol{q}}_t = \max(\boldsymbol{q}_t, 1)$ applied element-wise, resulting in normalized contact forces $\hat{\nabla} f(\boldsymbol{s}_t, \boldsymbol{a}_t) = \text{diag}(\hat{\boldsymbol{q}}_t) \nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)$. *Notably, this allows using a uniform contact threshold C across different tasks.*

Furthermore, considering that contact approximation forces are calculated separately in differentiable simulators, there is no need to use the full dynamics Jacobian. Instead, we employ the Jacobian, derived solely from contact forces. The differences between the SHAC and AHAC algorithms are comprehensively delineated in Appendix C.

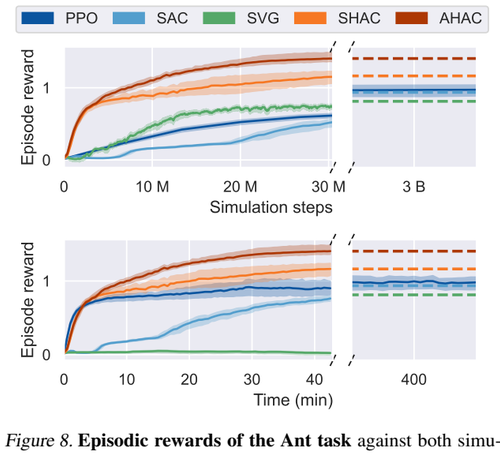


Figure 8. **Episodic rewards of the Ant task** against both simulation steps and wall clock time. The episodic reward is normalized by the highest mean reward achieved by PPO (i.e., PPO-normalized). The dashed lines represent the reward achieved by each respective algorithm at the end of their training runs.

5. Experiments

The objectives of this section are to (1) assess AHAC's ability to obtain higher asymptotic reward than MFRL baselines; (2) its efficiency in terms of wall-clock time and scalability to high-dimensional environments, and (3) identify the key contributing components of AHAC.

Setup. We evaluate the proposed approach, AHAC, across a set of 5 contemporary locomotion tasks, ranging from the simpler *Hopper* with $n = 11$ and $m = 3$, to the more complex *SNU Humanoid*, which features a muscle-actuated humanoid lower body with $m = 152$ (Figure 7). All tasks aim to maximize forward velocity, chosen for its benchmark relevance (Tassa et al., 2018) and complex optimization landscape as alluded to by previous results (Haarnoja et al., 2018; Hafner et al., 2019a). Experiments are based on *dflex*, a differentiable rigid-body simulator with soft contact approximation, introduced by (Xu et al., 2022), illustrated in Figure 7 and described in more detail in Appendix E. As is customary in prior work in empirical Deep RL (Tassa et al., 2018), we provide experimental results in an infinite-horizon setting and relax Assumption 2.1.

Metrics. We adopt statistical measures for a robust evaluation across 10 random seeds, utilizing the 50% Interquartile Mean (IQM) and 95% Confidence Interval (CI) as recommended for mitigating statistical uncertainties as suggested by (Agarwal et al., 2021). We also report absolute as well as normalized asymptotic rewards in Appendix H.

Baselines. This study compares first-order methods against zeroth-order methods. As such, we compare with state-of-the-art model-free methods, PPO (on-policy) (Schulman et al., 2017), and SAC (off-policy) (Haarnoja et al., 2018). For a comprehensive study, we also compare to SVG, a FO-MBRL method that does not utilize a differentiable simulator but instead learns the dynamics model(Amos et al., 2021). Additionally, we also compare our results to SHAC (Xu et al., 2022), one of the best-performing methods based on differentiable simulation. We refer the reader to SHAC (Xu et al., 2022) for additional comparisons with other model-based methods, which SHAC already outperforms. We have tuned all baselines individually to perform well per task and trained them until convergence. Due to long training times, we could not tune SVG since it was not vectorized and instead utilized the hyper-parameters from (Amos et al., 2021). All hyper-parameters are included in Appendix G.



图 7。**实验环境**。按动作空间 (m) 维度从左到右递增排列的运动任务：跳跃者 ($m = 3$)、蚂蚁 ($m = 8$)、Anymal ($m = 12$)、Humanoid ($m = 21$) 和 SNU 人形机器人 ($m = 152$)。

在实践中，仅基于动力学梯度 $\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)$ 进行截断被证明具有局限性，这是由于不同任务间接触力的变异性及其在学习过程中的演化所致。为解决这一问题，我们引入了一种针对接触力的归一化方法，利用每个状态维度上的修正加速度 $\hat{\boldsymbol{q}}_t = \max(\boldsymbol{q}_t, 1)$ 进行逐元素应用，从而得到归一化接触力 $\hat{\nabla} f(\boldsymbol{s}_t, \boldsymbol{a}_t) = \text{diag}(\hat{\boldsymbol{q}}_t) \nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)$。值得注意的是，这使得可以在不同任务中使用统一接触阈值 C。

此外，考虑到接触近似力在可微分仿真器中是单独计算的，因此无需使用完整的动力学雅可比矩阵。相反，我们采用仅由接触力推导出的雅可比矩阵。SHAC 与 AHAC 算法之间的差异在附录C中进行了全面阐述。

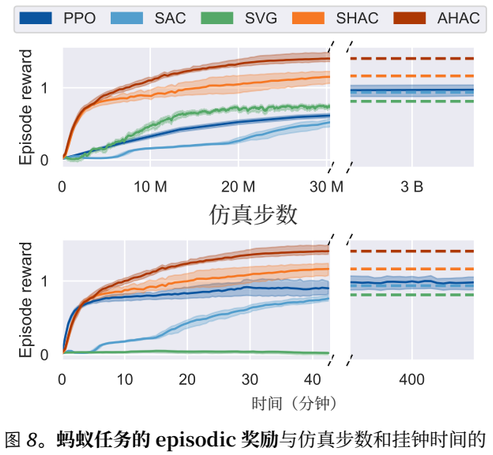


图 8。**蚂蚁任务的 episodic 奖励**与仿真步数和挂钟时间的关系。episodic 奖励已归一化为 PPO 所达到的最高平均奖励（即 PPO 归一化）。虚线表示各算法在训练结束时所达到的奖励。

5. 实验

本节的目标是：(1) 评估 AHAC 获得比免模型强化学习基线更高渐近奖励的能力；(2) 其在挂钟时间方面的效率以及对高维环境的可扩展性；以及 (3) 识别 AHAC 的关键贡献组件。

实验设置。我们在 5 项当代运动任务中评估了所提出的方法 AHAC，这些任务范围从较简单的具有 $n = 11$ 和 $m = 3$的跳跃者，到更复杂的 SNU 人形机器人，后者具有带有 $m = 152$ 的肌肉驱动人形机器人下肢（图 7）。所有任务旨在最大化前进速度，选择该指标是因为其基准相关性（Tassa 等，2018）以及如先前结果所述（Haarnoja 等，2018；Hafner 等人，2019a）所具有的复杂优化景观。实验基于 dflex，这是一种由 Xu 等（2022）引入的可微分刚体仿真器，具备软接触近似功能，如图 7 所示，并在附录 E 中有更详细的描述。按照以往深度强化学习实证研究中的惯例（Tassa 等，2018），我们在无限时域设定下提供实验结果，并放宽假设 2.1。

指标。我们采用统计度量，基于 10 个随机种子进行稳健评估，并使用 50% 四分位距
均值（IQM）及 95% 置信区间（CI），如建议用于缓解统计不确定性所推荐的那样——

由（Agarwal 等人，2021）提出。我们还在附录H中报告了绝对以及归一化的渐近奖励。

基线。本研究将一阶方法与零阶方法进行了对比。因此，我们与最先进的无模型方法进行了比较，包括 PPO（同策略）（Schulman 等，2017）和 SAC（异策略）（Haarnoja 等，2018）。为了进行全面研究，我们还与 SVG 进行了比较，这是一种不使用可微分仿真器而是学习动力学模型的一阶基于模型的强化学习方法（Amos 等，2021）。此外，我们还将我们的结果与 SHAC（Xu 等，2022）进行了比较，SHAC 是基于可微分仿真的表现最佳的方法之一。关于与其他基于模型的方法的更多比较（SHAC 已优于这些方法），我们请读者参考 SHAC（Xu 等，2022）。我们已对所有基线分别进行了调优，使其在每个任务中均表现良好，并训练至收敛。由于训练时间较长，我们无法对未向量化的 SVG 进行调优，而是采用了（Amos 等，2021）中的超参数。所有超参数均包含在附录G中。

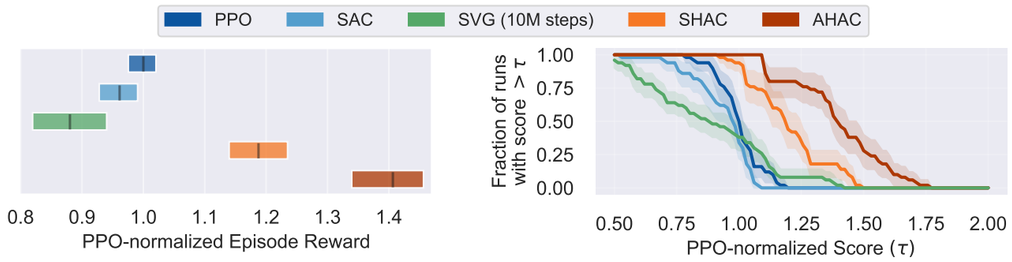


Figure 9. **Aggregate asymptotic statistics across all tasks.** The **left** figure shows 50% IQM with 95% CI of asymptotic episode rewards across 10 runs. We observe that AHAC is able to achieve 40% higher reward than our best MFRL baseline, PPO. The **right** figure shows score distributions as suggested by (Agarwal et al., 2021), which lets us understand the performance variability of each approach. Our proposed approach, AHAC, outperforms baselines even at the worst case, underlining the benefits of first-order methods.

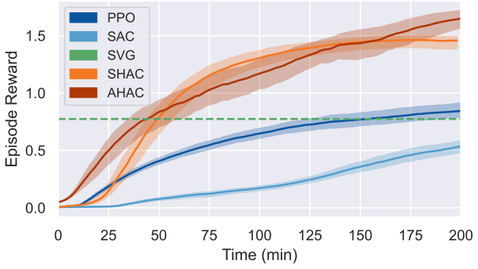


Figure 10. **Episodic rewards of the SNU Humanoid task**, a muscle-actuated humanoid lower body with $m = 152$. Results are smoothed using EWMA with $\alpha = 0.9$. We observe that both SHAC and AHAC scale better to high-dimensional tasks, with the latter achieving 61% more reward than PPO.

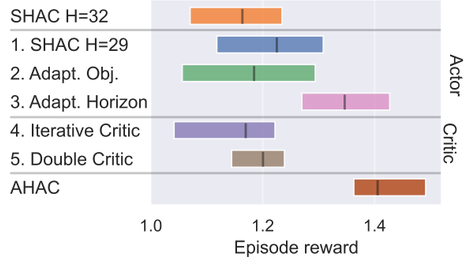


Figure 11. **Ablations of AHAC on the Ant task.** Ablating all additional introduced components reveals that the adaptive horizon objective contributes the most to improving episodic reward, while the double critic helps reduce run-to-run variance.

For comprehension, all rewards presented in this section are normalized by the maximum reward achieved by PPO per task. We include the raw numerical results in Appendix H along with further experiment details.

Results. First, we investigate the asymptotic performance of our method on the Ant task, a quadruped with symmetrical legs, $n = 37$ and $m = 8$. The results in Figure 8 show that AHAC achieves a 41% higher reward than the best model-free baseline, PPO, and also outperforms SHAC due to its gradient error avoidance technique. We acknowledge that MFRL methods are computationally simpler and thus also provide results against wall-clock time. Remarkably, PPO and SAC obtain worse episodic rewards over time compared to AHAC, even when they are trained for 3B timesteps, with $100\times$ more samples and $10\times$ more training time. These results suggest that even given practically infinite training data, MFRL methods cannot find truly optimal solutions due to the high variance of zeroth-order gradients.

This trend persists across all evaluated tasks, with AHAC consistently outperforming MFRL baselines. The aggregated statistics in Figure 9 suggest that *AHAC obtains a 40% higher reward than our main baseline, PPO*. Notably, in high-dimensional tasks such as the SNU Humanoid, AHAC's advantage becomes even more pronounced. Results from Figure 10 suggest that FO-MBRL methods significantly outperform MFRL baselines, with SHAC and AHAC obtaining 44% and 64% more reward than PPO, respectively. However, the larger confidence intervals for SHAC and AHAC hint at ongoing challenges with gradient variance associated with long rollouts. The score distributions in Figure 9 indicate that even with the higher variability, AHAC still outperforms MFRL baselines and exhibits better worst-case performance than SHAC. Additional results and raw metrics are provided in Appendix H.

Ablation study. To elucidate the performance improvements attributed to AHAC, we dissect its pivotal modifications, outlined in Appendix C. Beginning with the SHAC baseline set at $H = 32$, each ablation incrementally introduces a single modification:

1. SHAC H=29: using the H converged by AHAC.
2. Adapt. Obj.: SHAC with Eq. 10 and fixed $H = 32$.
3. Adapt. Horizon: SHAC with Eq. 10 and adapting H.
4. Iterative critic: SHAC with iterative critic training.
5. Double critic: SHAC with a double critic.

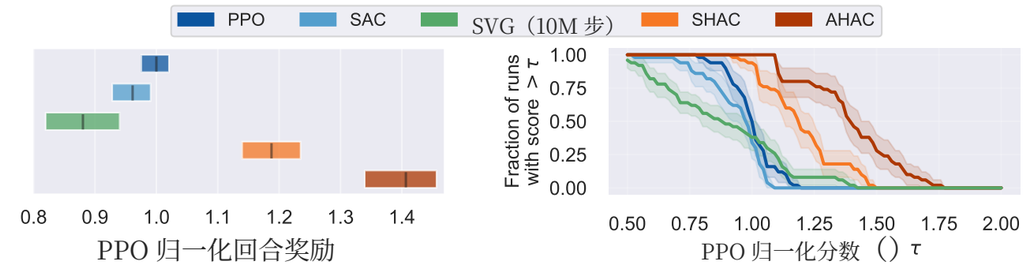


图 *9*。**所有任务的聚合渐近统计信息**。左**侧**图表展示了 10 次运行中渐近回合奖励的 50% IQM 及 95% CI。我们观察到，AHAC 能够实现比我们最佳的 MFRL 基线 PPO 高出 40% 的奖励。右**侧**图表展示了由（Agarwal 等人，2021）建议的分数分布，这使我们能够理解每种方法的性能变异性。我们提出的方法 AHAC 即使在最坏情况下也优于基线，凸显了一阶方法的优势。

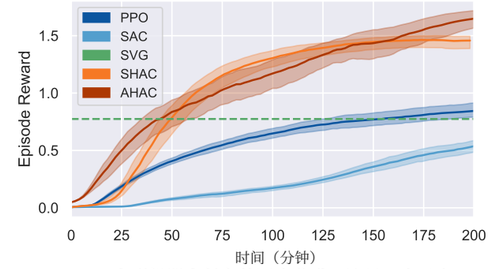


图 *10*。**SNU 人形机器人任务的回合奖励**，这是一个具有 $m = 152$的肌肉驱动人形机器人下肢。结果使用 EWMA 进行平滑处理，其中 $\alpha = 0.9$。我们观察到，SHAC 和 AHAC 都能更好地扩展到高维任务，其中后者实现的奖励比 PPO 高出 61%。

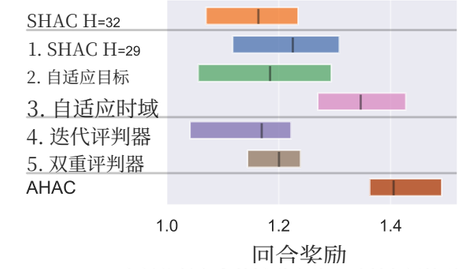


图 *11*.**AHAC 在蚂蚁任务上的消融实验**。对所有额外引入的组件进行消融实验表明，自适应时域目标对提升 episodic 奖励的贡献最大，而双重评判器有助于降低运行间方差。

为便于理解，本节中呈现的所有奖励均已按每个任务中 PPO 达到的最大奖励进行了归一化。我们在附录 H中提供了原始数值结果以及进一步的实验细节。

结果。首先，我们研究了我们的方法在 Ant 任务上的渐近性能，Ant 是一种具有对称腿的四足机器人，$n = 37$ 且 $m = 8$。图8中的结果表明，AHAC 实现的奖励比最佳的无模型基线 PPO 高出 41%，并且由于其梯度误差规避技术，其表现也优于 SHAC。我们承认 MFRL 方法在计算上更简单，因此也提供了针对挂钟时间的结果。值得注意的是，即使经过 30 亿时间步的训练，PPO 和 SAC 随时间推移获得的回合奖励仍不如 AHAC，尽管它们使用了$100\times$ 更多的样本和 $10\times$ 更多的训练时间。这些结果表明，即使给定实际上无限的训练数据，由于零阶梯度的高方差，MFRL 方法也无法找到真正的最优解。

这一趋势在所有评估任务中均持续存在，AHAC 始终优于免模型强化学习基线。图9 中的汇总统计数据表明，*AHAC* 获得的奖励比我们的主要基线 *PPO* 高出 *40%*。值得注意的是，在 SNU 人形机器人等高维任务中，AHAC 的优势更为显著。图10 的结果表明，FO-MBRL 方法显著优于免模型强化学习基线，其中 SHAC 和 AHAC 获得的奖励分别比 PPO 高出 44% 和 64%。然而，SHAC 和 AHAC 较大的置信区间暗示了与长 rollout 相关的梯度方差仍存在挑战。图9中的分数分布表明，即使存在较高的变异性，AHAC 仍然优于免模型强化学习基线，并且表现出比 SHAC 更好的最坏情况性能。其他结果和原始指标见附录H。

消融研究。为了阐明归因于 AHAC 的性能提升，我们剖析了其关键修改，详见附录C。从设置为 $H = 32$的 SHAC 基线开始，每次消融实验逐步引入单一修改：

1. SHAC H=29：使用由 AHAC 收敛的 H。
2. 自适应目标：SHAC 结合公式 10 并固定 $H = 32$。
3. 自适应时域：SHAC 结合公式 10 并自适应调整 H。
4. 迭代评判器：采用迭代评判器训练的 SHAC。
5. 双重评判器：采用双重评判器的 SHAC。

Algorithm	Policy Learning	Value Learning	Dynamics Model
PPO (Schulman et al., 2017)	ZOBG	Model-Free	-
SAC (Haarnoja et al., 2018)	0-step FOBG	Model-Free	-
MVE (Feinberg et al., 2018)	0-step FOBG	Model-Based	Deterministic NN
MBPO (Janner et al., 2019)	0-step FOBG	Model-Free	Ensemble NN
PIPPS (Parmas et al., 2018)	ZOBG & FOBG	-	Probabilistic NN
Dreamer (Hafner et al., 2019a)	FOBG	Model-Based	Probabilistic NN
IVG (Byravan et al., 2020)	FOBG	Model-Free	Deterministic NN
SVG (Amos et al., 2021)	FOBG	Model-Free	Deterministic NN
SHAC (Xu et al., 2022)	FOBG	Model-Free	Differentiable sim.
AHAC (ours)	FOBG	Model-Free	Differentiable sim.

Table 1. **Comparison between RL algorithms for continuous control.** We classify methods by the policy(actor) learning approach. ZOBG stands for methods using Zeroth-Order Batch Gradients following Eq. 2, while FOBG stands for First-Order Batch Gradient methods that differentiate through trajectories following Eq. 3. *Model-Based Value Learning* refers to methods leveraging Model-Based Value Expansion (MVE) (Feinberg et al., 2018), whereas *Model-Free critic learning* refers to methods using variants of TD(λ) (Sutton & Barto, 2018).

All experiments used the SHAC hyper-parameters, with the exception of the horizon learning rate α_ϕ, specifically adjusted for AHAC. Notably, SHAC with an adaptive horizon (3) is equivalent to AHAC without iterative critic training and single critic implementation. Results, depicted in Figure 11, reveal that incorporating an adaptive horizon significantly enhances the asymptotic reward. Intriguingly, adjusting to $H = 29$ improves rewards over the baseline, yet does not reach the efficacy of the full adaptive horizon approach. *This suggests that a static optimal horizon, even if advantageous at policy convergence, may not be optimal during training, leading to local minima.* Moreover, the double critic model notably reduces run-to-run variance, surpassing the performance stability of SHAC's single target-critic approach. Additional insights and detailed ablation results are available in Appendix I.

6. Related work

This section reviews recent advancements in continuous control RL, adhering to the actor-critic framework (Konda & Tsitsiklis, 1999), where the critic appraises state-action pairs and the actor identifies optimal actions. We categorize the methods based on their policy training, value estimation, and use of a dynamics model.

In the absence of a known dynamics model, Model-Free Reinforcement Learning (MFRL) methods prevail, which enable learning of action distributions based on state information. Proximal Policy Optimization (PPO) (Schulman et al., 2017) is an on-policy method that utilizes ZOBGs (Eq. 2) and performs gradient updates using recent on-policy samples. Soft Actor-Critic (SAC) (Haarnoja et al., 2018) exemplifies off-policy methods which use a replay buffer to learn from any data and update the actor using 0-step FOBGs defined as $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \mathbb{E}_{\boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)}[\nabla_{\boldsymbol{\theta}} Q(\boldsymbol{s}, \boldsymbol{a})]$

Alternatively, Model-Based Reinforcement Learning (MBRL) methods incorporate a dynamics model to inform learning, either derived from data or assumed a priori. This model can be used to aid the critic's return estimates, which can still be trained model-free (Janner et al., 2019) or through back-propagatable simulated returns via Model-Based Value Expansion (MVE) (Feinberg et al., 2018). Actor training varies; it can be done using 0-step FOBGs augmented by model-generated data (Janner et al., 2019; Feinberg et al., 2018). Alternatively, other work back-propagates through the dynamics model (Hafner et al., 2019a; Byravan et al., 2020) using FOBGs (Eq. 3). (Parmas et al., 2018) also combine ZOBGs and FOBGs, attempting to harness the best of both. Key recent work is summarized in Table 1.

With the emergence of differentiable simulation, many studies (Hu et al., 2019b; Liang et al., 2019; Huang et al., 2021; Du et al., 2021) have explored FOBG optimization by back-propagating through a dynamics model which we refer to as Back-Propagation-Through-Time (BPTT). However, BPTT faces challenges in long episodic RL tasks due to unstable gradients. (Xu et al., 2022) introduces Short Horizon Actor-Critic (SHAC), improving stability through a model-free critic and short rollouts, achieving performance comparable to MFRL with enhanced sample efficiency.

7. Conclusion

Model-free RL (MFRL) approaches are valued for their simplicity, minimal assumptions, and impressive performance. Yet, our study reveals their limitations in complex continuous control tasks, where they achieve good but suboptimal solutions due to high gradient variance. Conversely, First-Order Model-Based RL (FO-MBRL) methods, which leverage efficient gradient propagation through dynamics, have historically lagged behind MFRL in performance.

算法	策略学习	价值学习	动力学模型
PPO(Schulman 等，2017)	ZOBG	无模型的	-
SAC（Haarnoja 等，2018）	0 步 FOBG	无模型的	-
MVE（Feinberg 等，2018）	0 步 FOBG	基于模型的	确定性神经网络
MBPO (Janner 等人，2019)	0 步 FOBG	无模型的	集成神经网络
PIPPS（Parmas 等人，2018）	ZOBG 与 FOBG	-	概率神经网络
Dreamer（Hafner 等人，2019a）	FOBG	基于模型的	概率神经网络
IVG（Byravan 等，2020）	FOBG	无模型的	确定性神经网络
SVG（Amos 等，2021）	FOBG	无模型的	确定性神经网络
SHAC（Xu 等，2022）	FOBG	无模型的	可微分仿真
AHAC（ours）	FOBG	无模型的	可微分仿真

表 1.连续控制强化学习算法的比较。我们根据策略（actor）学习方法对方法进行归类。ZOBG 代表遵循公式2使用零阶批量梯度的方法，而 FOBG 代表遵循公式3通过轨迹进行微分的一阶批量梯度方法。基于模型的价值学习指利用基于模型的价值扩展（MVE）的方法（Feinberg 等，2018），而无模型评论家学习指使用 TD() 变体的方法（Sutton & Barto，2018）。

所有实验均使用了 SHAC 超参数，唯独视界学习率 α_ϕ 除外，该参数专为 AHAC 进行了调整。值得注意的是，具有自适应视界的 SHAC (3) 等同于未进行迭代 Critic 训练且采用单评论家实现的 AHAC。如图11所示，结果表明，引入自适应视界显著提升了渐近奖励。有趣的是，将参数调整至 $H = 29$ 虽能改善奖励表现，优于基线，但尚未达到完整自适应视界方法的效能。这表明，静态最优视界即使在策略收敛时具有优势，在训练过程中也可能并非最优，从而导致陷入局部极小值。此外，双评论家模型显著降低了运行间方差，其性能稳定性超越了 SHAC 的单目标评论家方法。更多见解及详细的消融实验结果请参阅附录I。

6. 相关工作

本节回顾连续控制强化学习的最新进展，遵循演员 - 评论家框架（Konda 与 Tsitsiklis，1999），其中评论家评估状态 - 动作对，而演员识别最优动作。我们根据其策略训练、价值估计以及动力学模型的使用对这些方法进行分类。

在缺乏已知动力学模型的情况下，无模型强化学习（MFRL）方法占据主导地位，这些方法能够基于状态信息学习动作分布。近端策略优化（PPO）（Schulman 等，2017）是一种同策略方法，它利用 ZOBG（公式2）并使用近期的同策略样本执行梯度更新。软演员 - 评论家（SAC）（Haarnoja 等，2018）是异策略方法的代表，它使用经验回放缓冲区从任意数据中学习，并利用定义为 $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) := \mathbb{E}_{\boldsymbol{a}_h \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_h)}[\nabla_{\boldsymbol{\theta}} Q(\boldsymbol{s}, \boldsymbol{a})]$ 的 0 步 FOBG 来更新 Actor。

或者，基于模型的强化学习（MBRL）方法引入动力学模型以指导学习，该模型可以从数据中推导或预先假设。此模型可用于辅助 Critic 的回报估计，这些估计仍可通过无模型方式训练（Janner 等，2019），或通过基于模型的价值扩展（MVE）（Feinberg 等，2018）利用可反向传播的模拟回报进行训练。Actor 的训练方式各异；可以使用由模型生成数据增强的 0 步 FOBG 来完成（Janner 等，2019；Feinberg 等，2018）。另外，其他研究通过动力学模型进行反向传播（Hafner 等，2019a；Byravan 等，2020），使用的是 FOBG（公式3）。（Parmas 等，2018）还结合了 ZOBG 和 FOBG，试图兼取两者之长。关键的最新工作总结于表1中。

随着可微分仿真的出现，许多研究（Hu 等，2019b；Liang 等，2019；Huang 等，2021；Du 等，2021）探索了通过对动力学模型进行反向传播的 FOBG 优化，我们将此称为时间反向传播（BPTT）。然而，由于梯度不稳定，BPTT 在长回合制强化学习任务中面临挑战。（Xu 等，2022）提出了短轨迹展开演员 - 评论家（SHAC），通过无模型评论家和短轨迹展开提高了稳定性，在提升样本效率的同时实现了与无模型强化学习相当的性能。

7. 结论

无模型强化学习（MFRL）方法因其简单性、假设最少以及令人印象深刻的性能而备受推崇。然而，我们的研究揭示了其在复杂连续控制任务中的局限性：由于梯度方差较高，这些方法虽能获得不错的解，却往往次优。相比之下，一阶基于模型的强化学习（FO-MBRL）方法通过动力学实现高效的梯度传播，但其性能历史上一直落后于 MFRL。

In this work, we analyze this issue in differentiable simulation through the scope of bias and variance. We derive Lemma 3.2 bounding the observed sample error of first-order gradients relative to zeroth-order gradients, coming to the conclusion that the source of the issue is stiff contact and long horizon rollouts. Based on these insights, we propose the Adaptive Horizon Actor-Critic (AHAC), a new FO-MBRL approach that adapts its rollout horizon during training. Our experiments show that AHAC outperforms MFRL baselines by 40% in complex locomotion tasks, even when the latter are provided with 10^6 times more data. Furthermore, out method maintains competitive time efficiency and shows better scalability to higher-dimensional tasks.

While AHAC outperforms MFRL methods and makes the case for first-order policy learning, it also necessitates the development of differentiable simulators. As such, we admire the simple yet capable MFRL approaches. Our work suggests that future research should not only focus on refining algorithmic approaches for policy learning but also on enhancing simulator technologies to more effectively manage gradient error. Moreover, the practical application of policies trained in differentiable simulators to real-world robotics remains a challenge.

Impact Statement

This paper presents work whose goal is to advance the field of Reinforcement Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Acknowledgment

The authors (IG, AG) were support by Stephen Fleming Early Career Chair as well as gifts from Nuro and Ford Motor Company, NSERC Discovery Award. This research was supported in part through services from the Partnership for an Advanced Computing Environment (PACE) at the Georgia Institute of Technology, Atlanta, Georgia, USA.

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在本工作中，我们从偏差和方差的角度，在可微分仿真中分析了这一问题。我们推导了引理3.2，该引理对一阶梯度相对于零阶梯度的观测样本误差进行了界定，并得出结论：问题的根源在于刚性接触和长视界展开。基于这些见解，我们提出了自适应时域演员 - 评论家算法（AHAC），这是一种新的 FO-MBRL 方法，能够在训练过程中自适应地调整其展开视界。实验表明，在复杂的运动任务中，AHAC 的表现比免模型强化学习基线高出 40%，即使后者获得了 10^6 倍更多的数据。此外，我们的方法保持了具有竞争力的时间效率，并展现出对更高维任务更好的可扩展性。

虽然 AHAC 优于 MFRL 方法并论证了一阶策略学习的价值，但它也 necessitates 可微分仿真器的开发。因此，我们赞赏简单而强大的 MFRL 方法。我们的工作表明，未来的研究不仅应专注于改进策略学习的算法方法，还应致力于增强仿真器技术，以更有效地管理梯度误差。此外，将在可微分仿真器中训练的策略应用于真实世界机器人学仍然是一个挑战。

影响声明

本文提出的工作旨在推动强化学习领域的发展。我们的工作可能产生许多潜在的社会后果，但我们认为无需在此特别强调任何一项。

致谢

作者（IG, AG）得到了 Stephen Fleming 早期职业讲席的支持，以及来自 Nuro、福特汽车公司和 NSERC 发现奖的资助。本研究部分得到了佐治亚理工学院（位于美国佐治亚州亚特兰大）高级计算环境合作伙伴关系（PACE）提供的服务支持。

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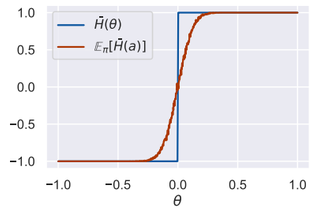
Tassa, Y., Doron, Y., Muldal, A., Erez, T., Li, Y., Casas, D. d. L., Budden, D., Abdolmaleki, A., Merel, J., Lefrancq, A., et al. Deepmind control suite. *arXiv preprint arXiv:1801.00690*, 2018.

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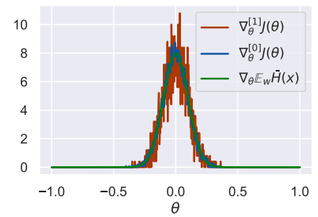
Xu, J., Chen, T., Zlokapa, L., Foshey, M., Matusik, W., Sueda, S., and Agrawal, P. An end-to-end differentiable framework for contact-aware robot design. *arXiv preprint arXiv:2107.07501*, 2021.

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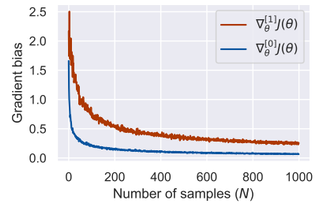
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(a) The Soft Heaviside function of Eq. 4.



(b) Gradient estimates at $N = 1000$.



(c) Gradient sample error for different sample sizes N. Estimated by comparing the difference between the gradient estimate and the true gradient below.

Figure 12. Gradient sample error study for the Soft Heaviside function shown in Eq. 4. Both ZOBG and FOBG exhibit sample errors at low sample sizes; however, FOBGs are especially susceptible to the "empirical bias" phenomena.

A. Heaviside example

This appendix provides additional details on the Heaviside example used to obtain intuition regarding FOBG sample error in Section 3.

$$\bar{H}(x) = \begin{cases} 1 & x > \nu/2 \\ 2x/\nu & |x| \leq \nu/2 \\ -1 & x < -\nu/2 \end{cases}$$

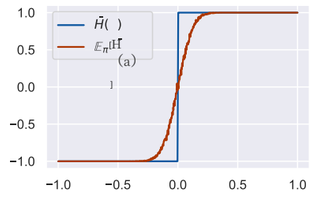
Under stochastic input $x \sim \pi_\theta(\cdot) = \theta + w$ where $w \sim \mathcal{N}(0, \sigma)$, we can obtain the expected value:

$$\begin{aligned}
\mathbb{E}_w\big[\bar{H}(x)\big] &= \int_{-\infty}^{\infty} \bar{H}(x)\pi_\theta(x)dx \\
&= -\int_{-\infty}^{-\nu/2} \pi_\theta(x)dx + \int_{-\nu/2}^{\nu/2} \frac{2x}{\nu}\pi_\theta(x)dx + \int_{\nu/2}^{\infty} \pi_\theta(x)dx \\
&= -\frac{1}{2}\operatorname{erfc}\left(\frac{\nu+2\theta}{2\sqrt{2}\nu}\right) + \frac{1}{2}\operatorname{erfc}\left(\frac{\nu-2\theta}{2\sqrt{2}\nu}\right) + \frac{\theta}{\nu}\operatorname{erf}\left(\frac{\nu-2\theta}{2\sqrt{2}\sigma}\right) + \frac{\theta}{\nu}\operatorname{erf}\left(\frac{\nu+2\theta}{2\sqrt{2}\sigma}\right) \\
&\quad + \frac{\sigma\sqrt{2}}{\nu\sqrt{\pi}}\Big(\exp -\frac{(\nu+2\theta)^2}{8\sigma^2} - \exp -\frac{(\nu-2\theta)^2}{8\sigma^2}\Big)
\end{aligned}$$

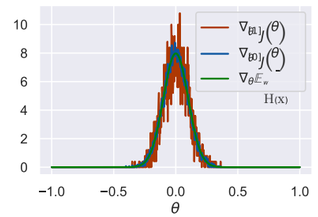
From the expectation, we can obtain the gradient w.r.t. the parameter of interest:

$$\begin{aligned}
\nabla_\theta \mathbb{E}_w\big[\bar{H}(x)\big] &= \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-(\nu+2\theta)^2}{8\sigma^2}\right) + \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-(\nu-2\theta)^2}{8\sigma^2}\right) + \frac{1}{\nu}\operatorname{erf}\left(\frac{\nu-2\theta}{2\sqrt{2}\sigma}\right) + \frac{1}{\nu}\operatorname{erf}\left(\frac{\nu+2\theta}{2\sqrt{2}\sigma}\right) \\
&\quad - \frac{\sqrt{2}\theta}{\sqrt{\pi}\nu\sigma}\exp\left(\frac{-(\nu-2\theta)^2}{8\sigma^2}\right) + \frac{\sqrt{2}\theta}{\sqrt{\pi}\nu\sigma}\exp\left(\frac{-(\nu+2\theta)^2}{8\sigma^2}\right) \\
&\quad - \frac{1}{\sqrt{2}\sigma\nu}\exp\big(-(\nu-2\theta)^{1/4\sigma^2}\big)(\nu-2\theta)^{1/4\sigma^2-1} - \frac{1}{\sqrt{2}\sigma\nu}\exp\big(-(\nu+2\theta)^{1/4\sigma^2}\big)(\nu+2\theta)^{1/4\sigma^2-1}
\end{aligned}$$

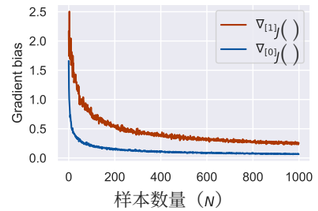
As seen from the equation above, the true gradient $\nabla_\theta \mathbb{E}_w\big[\bar{H}(x)\big] \neq 0$ at $\theta = 0$. However, using FOBG, we obtain $\nabla_\theta \bar{H}(a) = 0$ in samples where $|a| > \nu/2$, which occurs with probability at least $\nu/\sigma\sqrt{2\pi}$. Even though both ZOBG and FOBG are theoretically unbiased as $N \to \infty$, both exhibit "empirical bias", as shown in Figure 12.



(a) 式 4中的软 Heaviside 函数。



(b) $N = 1000$处的梯度估计。



(c) 不同样本量 N下的梯度样本误差。通过比较下方梯度估计与真实梯度之间的差异进行估算。

图 *12.*式 4所示软 Heaviside 函数的梯度样本误差研究。ZOBG 和 FOBG 在低样本量下均表现出样本误差；然而，FOBGs 尤其容易受到“经验偏差”现象的影响。

A. Heaviside 示例

本附录提供了关于 Heaviside 示例的更多细节，该示例用于在第 3 节中获得对 FOBG 采样误差的直观理解。

$$\bar{H}(x) = \begin{cases} 1 & x > \nu/2 \\ 2x/\nu & |x| \leq \nu/2 \\ -1 & x < -\nu/2 \end{cases}$$

在随机输入 $x \sim \pi_\theta(\cdot) = \theta + w$ （其中 $w \sim \mathcal{N}(0, \sigma)$）下，我们可以得到期望值：

$$\begin{aligned}
\mathbb{E}_w\big[\bar{H}(x)\big] &= \int_{-\infty}^{\infty} \bar{H}(x)\pi_\theta(x)dx \\
&= -\int_{-\infty}^{-\nu/2} \pi_\theta(x)dx + \int_{-\nu/2}^{\nu/2} \frac{2x}{\nu}\pi_\theta(x)dx + \int_{\nu/2}^{\infty} \pi_\theta(x)dx \\
&= -\frac{1}{2}\operatorname{erfc}\left(\frac{\nu+2\theta}{2\sqrt{2}\nu}\right) + \frac{1}{2}\operatorname{erfc}\left(\frac{\nu-2\theta}{2\sqrt{2}\nu}\right) + \frac{\theta}{\nu}\operatorname{erf}\left(\frac{\nu-2\theta}{2\sqrt{2}\sigma}\right) + \frac{\theta}{\nu}\operatorname{erf}\left(\frac{\nu+2\theta}{2\sqrt{2}\sigma}\right) \\
&\quad + \frac{\sigma\sqrt{2}}{\nu\sqrt{\pi}}\Big(\exp -\frac{(\nu+2\theta)^2}{8\sigma^2} - \exp -\frac{(\nu-2\theta)^2}{8\sigma^2}\Big)
\end{aligned}$$

根据该期望值，我们可以得到关于目标参数的梯度：

$$\begin{aligned}
\nabla_\theta \mathbb{E}_w\big[\bar{H}(x)\big] &= \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-(\nu+2\theta)^2}{8\sigma^2}\right) + \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-(\nu-2\theta)^2}{8\sigma^2}\right) + \frac{1}{\nu}\operatorname{erf}\left(\frac{\nu-2\theta}{2\sqrt{2}\sigma}\right) + \frac{1}{\nu}\operatorname{erf}\left(\frac{\nu+2\theta}{2\sqrt{2}\sigma}\right) \\
&\quad - \frac{\sqrt{2}\theta}{\sqrt{\pi}\nu\sigma}\exp\left(\frac{-(\nu-2\theta)^2}{8\sigma^2}\right) + \frac{\sqrt{2}\theta}{\sqrt{\pi}\nu\sigma}\exp\left(\frac{-(\nu+2\theta)^2}{8\sigma^2}\right) \\
&\quad - \frac{1}{\sqrt{2}\sigma\nu}\exp\big(-(\nu-2\theta)^{1/4\sigma^2}\big)(\nu-2\theta)^{1/4\sigma^2-1} - \frac{1}{\sqrt{2}\sigma\nu}\exp\big(-(\nu+2\theta)^{1/4\sigma^2}\big)(\nu+2\theta)^{1/4\sigma^2-1}
\end{aligned}$$

如上方公式所示，真实梯度 $\nabla_\theta \mathbb{E}_w\big[\bar{H}(x)\big] \neq 0$ at $\theta = 0$。然而，使用 FOBG 时，我们在$\nabla_\theta \bar{H}(a) = 0$ 样本中得到 $|a| > \nu/2$，这种情况发生的概率至少为 $\nu/\sigma\sqrt{2\pi}$。尽管 ZOBG 和 FOBG 在理论上都是无偏的（即 $N \to \infty$），但两者都表现出“经验偏差”，如图12所示。

B. Proof of Lemma 3.2

First we reiterate the assumptions to make this section self-sufficient and easier to read.

Assumption B.1. *Policy* $\pi(\cdot|s) : \mathbb{R}^n \times \mathbb{R}^d \to [0,1]^m$ *is continuously differentiable and Lipshitz smooth* $\|\nabla_{\boldsymbol{\theta}} \pi(a|s)\| \leq B_\pi$.

Assumption B.2. *Dynamics function* $f(s,a) : \mathbb{R}^n \times \mathbb{R}^d \to \mathbb{R}^n$ *is continuously differentiable and Lipshitz smooth in both arguments* $\|\nabla_s f(s,a) \leq B_f$ *and* $\|\nabla_a f(s,a)\| \leq B_f$.

Assumption B.3. *Reward function* $r(,a) : \mathbb{R}^n \times \mathbb{R}^d \to \mathbb{R}$ *is continuously differentiable and Lipshitz smooth* $\|\nabla_s r(s,a) \leq B_r$.

Assumption B.4. *ZOBG use baseline* b *subtraction which does not introduce gradient sample error* $\hat{\nabla}_\theta^{[}0](\theta) = \sum_{h=1}^{H} \big(r(s_h,a_h) - b\big)\nabla_\theta \log \pi_\theta(a_h|s_h)$

Proof. First, we expand our definition of sample error and define a random variable of a single Monte-Carlo sample

$$
\begin{aligned}
B = \left\| \mathbb{E}\left[\bar{\nabla}_{\boldsymbol{\theta}}^{[1]} J(\boldsymbol{\theta})\right] - \mathbb{E}\left[\bar{\nabla}_{\boldsymbol{\theta}}^{[0]} J(\boldsymbol{\theta})\right] \right\| &= \left\| \frac{1}{N}\sum_{i=1}^{N} \hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J_i(\boldsymbol{\theta}) - \frac{1}{N}\sum_{i=1}^{N} \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J_i(\boldsymbol{\theta}) \right\| \\
&= \frac{1}{N}\left\| \sum_{i=1}^{N} (\hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J_i(\boldsymbol{\theta}) - \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J_i(\boldsymbol{\theta})) \right\| \\
&\leq \|\hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J(\boldsymbol{\theta}) - \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J(\boldsymbol{\theta})\| \qquad (11)
\end{aligned}
$$

We drop the sample subscript i for simplicity and assume that $\boldsymbol{a} \sim \pi_{\boldsymbol{\theta}}(\cdot|\,\mathrm{sg}(\boldsymbol{s}))$ where $\mathrm{sg}(\cdot)$ is the stop-gradient operator makes the expansion of $\hat{\nabla}^{[1]} J(\boldsymbol{\theta})$ easier (Deng et al., 2024).

$$
\begin{aligned}
&\hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J(\boldsymbol{\theta}) - \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J(\boldsymbol{\theta}) \\
&= \nabla_{\boldsymbol{\theta}} \sum_{h=1}^{H} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \sum_{h=1}^{H} \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - b\big)\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \\
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) + \left(\sum_{h'=1}^{h-1} \Big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \Big) \nabla_{a_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'})^T \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \right)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \\
&\quad - \sum_{h=1}^{H} \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - b\big)\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \\
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \Big(\nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - b\big)\pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^{\circ -1} \Big) + \\
&\quad \left(\sum_{h'=1}^{h-1} \Big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \Big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \right)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \qquad (12)
\end{aligned}
$$

where $\mathbf{x}^{\circ -1}$ is the Hadamard inverse. Setting $b = r(\boldsymbol{s}_h, 0)$, we can exploit the Lipshitz smoothness of r. In general, for any function $f(\boldsymbol{x})$:

$$
\begin{aligned}
f(\boldsymbol{y}) &\leq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{L}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2 \\
f(0) &\leq f(\boldsymbol{x}) - \nabla f(\boldsymbol{x})^T \boldsymbol{x} + \frac{L}{2}\|\boldsymbol{x}\|^2 \\
f(\boldsymbol{x}) - f(0) &\geq \nabla f(\boldsymbol{x})^T \boldsymbol{x} - \frac{L}{2}\|\boldsymbol{x}\|^2 \\
f(\boldsymbol{x}) - f(0) &\geq (\nabla f(\boldsymbol{x})^T - \frac{L}{2}\boldsymbol{x}^T)\boldsymbol{x} \qquad (13)
\end{aligned}
$$

B. 引理证明 3.2

首先，我们重申这些假设，以使本节内容自洽且更易于阅读。

假设 B.1。 策略 $\pi(\cdot|s) : \mathbb{R}^n \times \mathbb{R}^d \to [0,1]^m$ 连续可微且利普希茨光滑 $\|\nabla_{\boldsymbol{\theta}} \pi(a|s)\| \leq B_\pi$。**假设 B.2**。 动力学函数 $f(s,a) : \mathbb{R}^n \times \mathbb{R}^d \to \mathbb{R}^n$ 在两个参数上均连续可微且利普希茨光滑 $\|\nabla_s f(s,a) \leq B_f$ *and* $\|\nabla_a f(s,a)\| \leq B_f$。

假设 B.3。 奖励函数 $r(,a) : \mathbb{R}^n \times \mathbb{R}^d \to \mathbb{R}$ 连续可微且利普希茨光滑 $\|\nabla_s r(s,a) \leq B_r$。

假设 B.4。 *ZOBG* 使用基线 b 减法，不会引入梯度样本误差 $\hat{\nabla}_\theta^{[}0](\theta) = \sum_{h=1}^{H} \big(r(s_h,a_h) - b\big)\nabla_\theta \log \pi_\theta(a_h|s_h)$

证明。首先，我们扩展样本误差的定义，并定义单个蒙特卡洛采样的随机变量

$$
\begin{aligned}
B = \left\| \mathbb{E}\left[\bar{\nabla}_{\boldsymbol{\theta}}^{[1]} J(\boldsymbol{\theta})\right] - \mathbb{E}\left[\bar{\nabla}_{\boldsymbol{\theta}}^{[0]} J(\boldsymbol{\theta})\right] \right\| &= \left\| \frac{1}{N}\sum_{i=1}^{N} \hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J_i(\boldsymbol{\theta}) - \frac{1}{N}\sum_{i=1}^{N} \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J_i(\boldsymbol{\theta}) \right\| \\
&= \frac{1}{N}\left\| \sum_{i=1}^{N} (\hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J_i(\boldsymbol{\theta}) - \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J_i(\boldsymbol{\theta})) \right\| \\
&\leq \|\hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J(\boldsymbol{\theta}) - \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J(\boldsymbol{\theta})\| \qquad (11)
\end{aligned}
$$

为简化起见，我们省略样本下标 i 并假设 $\boldsymbol{a} \sim \pi_{\boldsymbol{\theta}}(\cdot|\,\mathrm{sg}(\boldsymbol{s}))$ 其中$\mathrm{sg}(\cdot)$ 是梯度停止算子，使得 $\hat{\nabla}^{[1]} J(\boldsymbol{\theta})$ 的展开更容易（Deng 等，2024）。

$$
\begin{aligned}
&\hat{\nabla}_{\boldsymbol{\theta}}^{[1]} J(\boldsymbol{\theta}) - \hat{\nabla}_{\boldsymbol{\theta}}^{[0]} J(\boldsymbol{\theta}) \\
&= \nabla_{\boldsymbol{\theta}} \sum_{h=1}^{H} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \sum_{h=1}^{H} \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - b\big)\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \\
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) + \left(\sum_{h'=1}^{h-1} \Big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \Big) \nabla_{a_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'})^T \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \right)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \\
&\quad - \sum_{h=1}^{H} \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - b\big)\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \\
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \Big(\nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - b\big)\pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^{\circ -1} \Big) + \\
&\quad \left(\sum_{h'=1}^{h-1} \Big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \Big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \right)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \qquad (12)
\end{aligned}
$$

其中 $\mathbf{x}^{\circ -1}$ 是哈达玛逆。设 $b = r(\boldsymbol{s}_h, 0)$，我们可以利用 r 的利普希茨光滑性。一般而言，对于任意函数 $f(\boldsymbol{x})$：

$$
\begin{aligned}
f(\boldsymbol{y}) &\leq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{L}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2 \\
f(0) &\leq f(\boldsymbol{x}) - \nabla f(\boldsymbol{x})^T \boldsymbol{x} + \frac{L}{2}\|\boldsymbol{x}\|^2 \\
f(\boldsymbol{x}) - f(0) &\geq \nabla f(\boldsymbol{x})^T \boldsymbol{x} - \frac{L}{2}\|\boldsymbol{x}\|^2 \\
f(\boldsymbol{x}) - f(0) &\geq (\nabla f(\boldsymbol{x})^T - \frac{L}{2}\boldsymbol{x}^T)\boldsymbol{x} \qquad (13)
\end{aligned}
$$

Applying $b = r(\boldsymbol{s}_h, 0)$ to Eq. 12 yields

$$
\begin{aligned}
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \Big(\nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - r(\boldsymbol{s}_h, 0)\big) \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^{\circ -1} \Big) + \\
&\quad \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \\
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \big(\nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \frac{B_r}{2} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \big) \\
&\quad + \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \\
&= \frac{B_r}{2} \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) + \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h)
\end{aligned} \tag{14}
$$

Plug Eq. 14 into Eq. 11:

$$
\begin{aligned}
B &\leq \left\| \hat{\nabla}^{[1]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) - \hat{\nabla}^{[0]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \right\| \\
&= \left\| \frac{B_r}{2} \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) + \Big(\sum_{h'=1}^{h-1} \prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \right\| \\
&\qquad \text{Apply Eq. 13} \\
&\leq \left\| \frac{B_r}{2} \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \right\| \\
&\qquad + \left\| \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \right\| \\
&\qquad \text{since } \|\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\| \leq B_\pi \text{ and } \|\pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\| \leq 1 \\
&\leq \frac{1}{2} H B_r B_\pi + \left\| \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \right\| \\
&\leq \frac{1}{2} H B_r B_\pi + (H-1) B_r B_\pi B_f^{H-1} \\
&\leq \frac{1}{2} H B_r B_\pi + H B_r B_\pi B_f^{H-1} \\
&= H B_r B_\pi (\frac{1}{2} + B_f^{H-1})
\end{aligned}
$$

□

将 $b = r(\boldsymbol{s}_h, 0)$ 应用于公式 12 可得

$$
\begin{aligned}
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \Big(\nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \big(r(\boldsymbol{s}_h, \boldsymbol{a}_h) - r(\boldsymbol{s}_h, 0)\big) \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^{\circ -1} \Big) + \\
&\quad \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \\
&= \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \big(\nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) - \nabla_{\boldsymbol{a}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \frac{B_r}{2} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \big) \\
&\quad + \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \\
&= \frac{B_r}{2} \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) + \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h)
\end{aligned} \tag{14}
$$

将式 14代入式 11:

$$
\begin{aligned}
B &\leq \left\| \hat{\nabla}^{[1]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) - \hat{\nabla}^{[0]}_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \right\| \\
&= \left\| \frac{B_r}{2} \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) + \Big(\sum_{h'=1}^{h-1} \prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \right\| \\
&\qquad \text{Apply Eq. 13} \\
&\leq \left\| \frac{B_r}{2} \sum_{h=1}^{H} \nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)^T \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h) \right\| \\
&\qquad + \left\| \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \right\| \\
&\qquad \text{since } \|\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\| \leq B_\pi \text{ and } \|\pi_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\| \leq 1 \\
&\leq \frac{1}{2} H B_r B_\pi + \left\| \Big(\sum_{h'=1}^{h-1} \big(\prod_{t=h'+1}^{h-1} \nabla_{\boldsymbol{s}_t} f(\boldsymbol{s}_t, \boldsymbol{a}_t) \big) \nabla_{\boldsymbol{a}_{h'}} f(\boldsymbol{s}_{h'}, \boldsymbol{a}_{h'}) \nabla_{\boldsymbol{\theta}} \pi(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'}) \Big)^T \nabla_{\boldsymbol{s}_h} r(\boldsymbol{s}_h, \boldsymbol{a}_h) \right\| \\
&\leq \frac{1}{2} H B_r B_\pi + (H-1) B_r B_\pi B_f^{H-1} \\
&\leq \frac{1}{2} H B_r B_\pi + H B_r B_\pi B_f^{H-1} \\
&= H B_r B_\pi (\frac{1}{2} + B_f^{H-1})
\end{aligned}
$$

□

C. Summary of modifications

To develop the Adaptive Horizon Actor-Critic (AHAC) algorithm, we used the Short Horizon Actor-Critic (SHAC) algorithm (Xu et al., 2022) as a starting point. This section details all modifications applied to the SHAC in order to derive AHAC and achieve the reported results in this paper. We also note that some of these are not exclusive to either approach approach.

1. **Adaptive horizon objective** - instead of optimizing the short horizon rollout return, we introduce the new constrained objective shown in Equation 9. To optimise that and adapt the horizon H, we introduced the dual problem in Equation 10 and optimised it directly for policy parameters $\boldsymbol{\theta}$ and the Lagrangian coefficients ϕ.

$$\underbrace{\underbrace{J(\boldsymbol{\theta}) := \sum_{h=t}^{t+T-1} \gamma^{h-t} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \gamma^t V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+T})}_{\text{SHAC objective}} \quad s.t. \quad \|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| \leq C \quad \forall t \in \{0, .., T\}}_{\text{AHAC objective}}$$

2. **Double critic** - the original implementation of SHAC struggled with more complex tasks such as Humanoid due to its highly non-convex value landscape. The authors of (Xu et al., 2022) solved that by introducing a delayed target critic similar to prior work in deep RL (Lillicrap et al., 2015). We found that approach brittle and required more hyper-parameter tuning. Instead, we replaced it with a double critic similar to SAC (Haarnoja et al., 2018). For our work, we found that it reduced the variance of asymptotic rewards achieved by AHAC while removing a hyperparameter. While this technique is usually applied to off-policy algorithms, we find it helpful in highly parallelized simulations due to the high data throughput.

3. **Critic training until convergence** - empirically we found that different problems present different value landscapes. The more complex the landscape, the more training the critic required and the critic often failed to fit the data with the limited number of critic iterations done in SHAC (16). Instead of training the critic for a fixed number of iterations, we trained the (dual) critic of AHAC until convergence defined by $\sum_{i=n-5}^{n} \mathcal{L}_i(\psi) - \mathcal{L}_{i-1}(\psi) < 0.5$ where $\mathcal{L}_i(\psi)$ is the critic loss for mini-batch iteration i. We allowed the critic to be trained for a maximum of 64 iterations. We found that this resulted in asymptotic performance improvements on more complex tasks such as Humanoid and SNU Humanoid, while removing yet another hyper-parameter.

D. AHAC-1 algorithm

Algorithm 2 shows the single-environment version of AHAC that was described in Section 4.1. While this algorithm applies the contact truncation technique perfectly and avoids all stiff contact, it is also not vectorizable. When attempting to vectorize AHAC-1, it necessitates cutting off compute graphs per-environment based on the individual environment dynamics. This is impossible to accomplish with typical deep learning frameworks such as PyTorch. Another alternative would be to execute different environments in different threads, but unfortunately, that does not benefit from GPU acceleration.

E. Differentiable Simulation Setup: dflex

The experimental simulator, *dflex* (Xu et al., 2022), employed in Section 5, is a GPU-based differentiable simulator utilizing the Featherstone formulation for forward dynamics and a spring-damper contact model with Coulomb friction.

The dynamics function f is modeled by solving the forward dynamics equations:

$$M\ddot{q} = J^T \mathcal{F}(q, \dot{q}) + c(q, \dot{q}) + \tau(q, \dot{q}, a)$$

where, $q, \dot{q}, \ddot{q}$ are joint coordinates, velocities, and accelerations, respectively. $\mathcal{F}$ represents external forces, c includes Coriolis forces, and τ denotes joint-space actuation. Mass matrix M and Jacobian J are computed concurrently using one thread per-environment. The composite rigid body algorithm (CRBA) is employed for articulation dynamics, enabling caching of the matrix factorization for reuse in the backward pass through parallel Cholesky decomposition.

After determining joint accelerations $\ddot{q}$, a semi-implicit Euler integration step updates the system state $s = (q, \dot{q})$. Torque-based control is employed for simple environments, where the policy outputs τ at each timestep. For further details, see (Xu et al., 2022). It is noted that *dflex* is no longer actively developed and has been succeeded by *warp* (Macklin, 2022).

C. 修改总结

为了开发自适应时域演员 - 评论家算法（AHAC），我们以短视界演员 - 评论家（SHAC）算法（Xu 等，2022）为起点。本节详细介绍了为推导 AHAC 并实现本文报告结果而对 SHAC 应用的所有修改。我们还指出，其中一些修改并非任一方法所独有。

1. **自适应视界目标**- 我们不再优化短视界滚动回报，而是引入了公式9中所示的新约束目标。为了优化该目标并自适应地调整视界 H，我们在公式10 中引入对偶问题，并直接针对策略参数 $\boldsymbol{\theta}$ 和拉格朗日系数 ϕ对其进行优化。

$$\underbrace{\underbrace{J(\boldsymbol{\theta}) := \sum_{h=t}^{t+T-1} \gamma^{h-t} r(\boldsymbol{s}_h, \boldsymbol{a}_h) + \gamma^t V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+T})}_{\text{SHAC objective}} \quad s.t. \quad \|\nabla f(\boldsymbol{s}_t, \boldsymbol{a}_t)\| \leq C \quad \forall t \in \{0, .., T\}}_{\text{AHAC 目标}}$$

2. **双重评判器**- SHAC 的原始实现在处理 Humanoid 等更复杂任务时举步维艰，这是由于其高度非凸的值景观所致。（Xu 等，2022）的作者通过引入延迟目标评论家解决了这一问题，这与深度强化学习中的先前工作（Lillicrap 等，2015）类似。我们发现该方法脆弱且需要更多的超参数调优。因此，我们将其替换为类似于 SAC（Haarnoja 等，2018）的双重评判器。在我们的工作中，我们发现该方法在减少 AHAC 所实现的渐近奖励方差的同时，还省去了一个超参数。虽然该技术通常应用于异策略算法，但我们发现由于高数据吞吐量，它在高度并行化仿真中同样有益。

3. **评论家训练至收敛**——根据经验，我们发现不同的问题呈现出不同的值景观。景观越复杂，评论家所需的训练就越多，而且在 SHAC 中有限的评论家迭代次数（16 次）下，评论家往往无法拟合数据。我们不再为评论家训练固定的迭代次数，而是将 AHAC 的（Dual）评论家训练至收敛，收敛条件定义为 $\sum_{i=n-5}^{n} \mathcal{L}_i(\psi) - \mathcal{L}_{i-1}(\psi) < 0.5$，其中 $\mathcal{L}_i(\psi)$ 是小批量迭代 i的评论家损失。我们允许评论家最多训练 64 次迭代。我们发现，这在 Humanoid 和 SNU 人形机器人等更复杂的任务上带来了渐近性能提升，同时消除了另一个超参数。

D. AHAC-1 算法

算法2 展示了第4.1节中描述的 AHAC 单环境版本。虽然该算法完美地应用了接触截断技术并避免了所有刚性接触，但它不可矢量化。当尝试对 AHAC-1 进行矢量化时，需要根据独立环境动力学切断每个环境的计算图。使用 PyTorch 等典型的深度学习框架无法实现这一点。另一种替代方案是在不同的线程中执行不同的环境，但遗憾的是，这无法利用 GPU 加速。

E. 可微分仿真设置：dflex

第*5*节所采用的实验模拟器dflex（Xu 等，2022）是一个基于 GPU 的可微模拟器，它利用 Featherstone 公式进行前向动力学计算，并采用带有库仑摩擦的弹簧 - 阻尼接触模型。

动力学函数 f 通过求解前向动力学方程来建模：

$$M\ddot{q} = J^T \mathcal{F}(q, \dot{q}) + c(q, \dot{q}) + \tau(q, \dot{q}, a)$$

其中，q、$\dot{q}$、$\ddot{q}$ 分别为关节坐标、速度和加速度。$\mathcal{F}$ 表示外力，c 包含科里奥利力，τ 表示关节空间驱动。质量矩阵 M 和雅可比矩阵 J 通过每个环境一个线程的方式并发计算。复合刚体算法（CRBA）用于关节动力学，支持对矩阵分解进行缓存，以便在反向传播中通过并行 Cholesky 分解重复使用。

在确定关节加速度后 $\ddot{q}$，通过半隐式欧拉积分步更新系统状态 $s = (q, \dot{q})$。对于简单环境，采用基于力矩的控制，其中策略在每个时间步输出 τ。更多详情，请参阅（Xu 等，2022）。需要注意的是，*dflex* 已不再积极开发，并由*warp*（Macklin，2022）取代。

Algorithm 2 Adaptive Horizon Actor-Critic 1 (Single environment)

Given: γ: discount rate
Given: $\alpha_{\boldsymbol{\theta}}, \alpha_{\boldsymbol{\psi}}$: learning rates
Given: H: maximum trajectory length
Given: C: contact threshold
Initialize learnable parameters $\boldsymbol{\theta}, \boldsymbol{\psi}$
$t \leftarrow 0$

while episode not done **do**
  ▷ Rollout policy
  Initialize rollout buffer D
  while $\|\nabla f\| \leq C$ and $h \leq H$ **do**
    $\boldsymbol{a}_t \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_t)$
    $r_t = r(\boldsymbol{s}_t, \boldsymbol{a}_t)$
    $s_{t+1} = f(\boldsymbol{s}_t, \boldsymbol{a}_t)$
    $D \leftarrow D \cup \{(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h}, r_{t+h}, V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+h+1}))\}$
    $t \leftarrow t + 1$
  end while
  ▷ Train actor using Eq. 8
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$
  ▷Train critic using Eq. 7
  while not converged **do**
    sample $(\boldsymbol{s}, \hat{V}(\boldsymbol{s})) \sim D$
    $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} + \alpha_{\boldsymbol{\psi}} \nabla_{\boldsymbol{\psi}} \mathcal{L}(\boldsymbol{\psi})$
  end while
end while



Figure 13. Locomotion environments (left to right): Hopper, Ant, Anymal, Humanoid and SNU Humanoid.

F. Environment details

In this paper, we explore 5 locomotion tasks with increasing complexity. They are described below and shown in Figure 13.

1. **Hopper**, a single-legged robot jumping only in one axis with $n = 11$ and $m = 3$.
2. **Ant**, a four-legged robot with $n = 37$ and $m = 8$.
3. **Anymal**, a more sophisticated quadruped with $n = 49$ and $m = 12$ modeled after (Hutter et al., 2016).
4. **Humanoid**, a classic contact-rich environment with $n = 76$ and $m = 21$, which requires extensive exploration to find a good policy.
5. **SNU Humanoid**, a version of Humanoid lower body where instead of joint torque control, the robot is controlled via $m = 152$ muscles intended to challenge the scaling capabilities of algorithms.

All tasks share the same common main objective - maximize forward velocity v_x:

Algorithm 2 Adaptive Horizon Actor-Critic 1 (Single environment)

Given: γ: discount rate
Given: $\alpha_{\boldsymbol{\theta}}, \alpha_{\boldsymbol{\psi}}$: learning rates
Given: H: maximum trajectory length
Given: C: contact threshold
Initialize learnable parameters $\boldsymbol{\theta}, \boldsymbol{\psi}$
$t \leftarrow 0$

while episode not done **do**
  ▷ Rollout policy
  Initialize rollout buffer D
  while $\|\nabla f\| \leq C$ and $h \leq H$ **do**
    $\boldsymbol{a}_t \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s}_t)$
    $r_t = r(\boldsymbol{s}_t, \boldsymbol{a}_t)$
    $s_{t+1} = f(\boldsymbol{s}_t, \boldsymbol{a}_t)$
    $D \leftarrow D \cup \{(\boldsymbol{s}_{t+h}, \boldsymbol{a}_{t+h}, r_{t+h}, V_{\boldsymbol{\psi}}(\boldsymbol{s}_{t+h+1}))\}$
    $t \leftarrow t + 1$
  end while
  ▷ Train actor using Eq. 8
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$
  ▷Train critic using Eq. 7
  while not converged **do**
    sample $(\boldsymbol{s}, \hat{V}(\boldsymbol{s})) \sim D$
    $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} + \alpha_{\boldsymbol{\psi}} \nabla_{\boldsymbol{\psi}} \mathcal{L}(\boldsymbol{\psi})$
  end while
end while



图 *13*。运动环境（从左到右）：跳跃者、蚂蚁、Anymal、Humanoid 和 SNU 人形机器人。

F. 环境详情

在本文中，我们探索了 5 个复杂度递增的运动任务。它们如下所述，并展示于图13中。

1. **跳跃者**，一种仅沿单一轴跳跃的单腿机器人，具有 $n = 11$和 $m = 3$。
2. **蚂蚁**，一种具有 $n = 37$和 $m = 8$的四足机器人。
3. **Anymal**，一种更为复杂的四足机器人，具有 $n = 49$和 $m = 12$，其建模参考了（Hutter 等，2016）。
4. **Humanoid**，一个经典的丰富接触环境，包含 $n = 76$ 和 $m = 21$，需要大量探索才能找到良好的策略。
5. **SNU 人形机器人**，是 Humanoid 下半身的一个版本，它不使用关节力矩控制，而是通过$m = 152$肌肉进行控制，旨在挑战算法的扩展能力。

所有任务共享同一个共同的主要目标——最大化前进速度 v_x:

Environment	Reward
Hopper	$v_x + R_{height} + R_{angle} - 0.1\|\boldsymbol{a}\|_2^2$
Ant	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.01\|\boldsymbol{a}\|_2^2$
Anymal	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.01\|\boldsymbol{a}\|_2^2$
Humanoid	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.002\|\boldsymbol{a}\|_2^2$
Humanoid STU	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.002\|\boldsymbol{a}\|_2^2$

Table 2. Rewards used for each task benchmarked in Section 5

We additionally use auxiliary rewards R_{height} to incentivize the agent to, R_{angle} to keep the agent's normal vector point up, $R_{heading}$ to keep the agent's heading pointing towards the direction of running and a norm over the actions to incentivize energy-efficient policies. For most algorithms, none of these rewards, apart from the last one, are crucial to succeeding in the task. However, all of them aid learning policies faster.

$$R_{height} = \begin{cases} h - h_{term} & if h \geq h_{term} \\ -200(h - h_{term})^2 & if h < h_{term} \end{cases}$$

$$R_{angle} = 1 - \left(\frac{\theta}{\theta_{term}}\right)^2$$

$R_{angle} = \|\boldsymbol{q}_{forward} - \boldsymbol{q}_{agent}\|_2^2$ is the difference between the heading of the agent $\boldsymbol{q}_{agent}$ and the forward vector $\boldsymbol{q}_{agent}$. h is the height of the CoM of the agent and θ is the angle of its normal vector. h_{term} and θ_{term} are parameters that we set for each environment depending on the robot morphology. Similar to other high-performance RL applications in simulation, we find it crucial to terminate episode early if the agent exceeds these termination parameters. However, it is worth noting that AHAC is still capable of solving all tasks described in the paper without these termination conditions, albeit slower.

G. Hyper-parameters

This section details all hyper-parameters used in the main experiments in Section 5. PPO and SAC, as our MFRL baselines, have been tuned to perform well across all tasks, including task-specific hyper-parameters. SVG has not been specifically tuned for all benchmarks due to time limitations but instead uses the hyper-parameters presented in (Amos et al., 2021).[1] SHAC is tuned to perform well across all tasks using a fixed $H = 32$ as in the original work (Xu et al., 2022). AHAC shares all of its common hyper-parameters with SHAC and only has its horizon learning rate α_ϕ tuned per-task. The contact threshold C and iterative critic training criteria did not benefit from tuning. Note that the double critic employed by AHAC uses the same hyper-parameters used by the SHAC critic. Therefore, we have left AHAC under-tuned in comparison to SHAC in order to highlight the benefits of the adaptive horizon mechanism presented in this work.

Table 3 shows common hyper-parameters shared between all tasks. While table 4 shows hyper-parameters specific to each problem,. Where possible, we attempted to use the hyper-parameters suggested by the original works; however, we also attempted to share hyper-parameters between algorithms to ease comparison. If a specific hyperparameter is not mentioned, then it is the one used in the original work behind the specific algorithm.

[1]Tuning SVG proved difficult as we were unable to vectorize the algorithm, resulting in up to 2-week training times. This made it difficult to tune for our benchmarks

环境	奖励
跳跃者	$v_x + R_{height} + R_{angle} - 0.1\|\boldsymbol{a}\|_2^2$
Ant	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.01\|\boldsymbol{a}\|_2^2$
Anymal	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.01\|\boldsymbol{a}\|_2^2$
Humanoid	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.002\|\boldsymbol{a}\|_2^2$
人形机器人 STU	$v_x + R_{height} + 0.1R_{angle} + R_{heading} - 0.002\|\boldsymbol{a}\|_2^2$

表 2。第5节中基准测试的每项任务所使用的奖励

我们 additionally 使用辅助奖励 R_{height} 来激励智能体， R_{angle} 使智能体的法向量保持向上，$R_{heading}$ 使智能体的朝向指向奔跑方向，并对动作施加范数以激励节能策略。对于大多数算法而言，除了最后一项奖励外，这些奖励对于成功完成任务并非关键。然而，它们都有助于更快地学习策略。

$$R_{height} = \begin{cases} h - h_{term} & if h \geq h_{term} \\ -200(h - h_{term})^2 & if h < h_{term} \end{cases}$$

$$R_{angle} = 1 - \left(\frac{\theta}{\theta_{term}}\right)^2$$

$R_{angle} = \|\boldsymbol{q}_{forward} - \boldsymbol{q}_{agent}\|_2^2$ 是代理朝向 $\boldsymbol{q}_{agent}$ 与前向向量 $\boldsymbol{q}_{agent}$之间的差值。 h是代理质心（CoM）的高度，θ 是其法向量的角度。 h_{term} 和 θ_{term} 是我们根据机器人形态为每个环境设定的参数。与其他仿真中的高性能强化学习（RL）应用类似，我们发现如果代理超出这些终止参数，提前终止回合至关重要。然而，值得注意的是，即使没有这些终止条件，AHAC 仍然能够解决论文中描述的所有任务，尽管速度较慢。

G. 超参数

本节详细介绍了第5节主要实验中使用的所有超参数。PPO和SAC作为我们的免模型强化学习基线，已针对所有任务进行了调优，包括特定于任务的超参数。由于时间限制，SVG未针对所有基准测试进行专门调优，而是采用了（Amos 等，2021）中提出的超参数。[1]SHAC经过调优以在所有任务中表现良好，使用了固定的 $H = 32$，如同原始工作（Xu 等，2022）所示。AHAC与其共享所有通用超参数，仅对其视界学习率 α_ϕ 进行每项任务的单独调优。接触阈值 C 和迭代评论家训练准则并未从调优中获益。请注意，AHAC采用的双重评判器使用了与SHAC评判器相同的超参数。因此，为了突显本文提出的自适应视界机制的优势，我们有意让AHAC相较于SHAC处于欠调优状态。

表3 展示了所有任务共用的常见超参数。而表4 则展示了针对每个问题特定的超参数。在可能的情况下，我们尝试使用原始工作建议的超参数；同时，我们也尝试在不同算法间共享超参数以方便比较。如果未提及特定超参数，则采用该特定算法背后原始工作中所使用的超参数。

[1]调整 SVG 被证明很困难，因为我们无法将该算法向量化，导致训练时间长达 2 周。这使得我们难以针对基准测试进行调整

	AHAC	SHAC	PPO	SAC	SVG
Mini-epochs		16	5		4
Batch size	8	8	8	32	1024
λ	0.95	0.95	0.95		
γ	0.99	0.99	0.99	0.99	0.99
H - horizon		32	32		3
C - contact thresh.	500				
Grad norm	1.0	1.0	1.0		
ϵ			0.2		
Actor $log(\sigma)$ bounds				(-5,2)	(-5,2)
α - temperature				0.2	0.1
λ_α				10^{-4}	10^{-4}
$\|D\|$ - buffer size				10^6	10^6
Seed steps	0	0	0	10^4	10^4

Table 3. Table of hyper-parameters for all algorithms benchmarked in Section 5. These are shared across all tasks.

	Hopper	Ant	Anymal	Humanoid	SNU Humanoid
Actor layers	(128, 64, 32)	(128, 64, 32)	(256, 128)	(256, 128)	(512, 256)
Actor $\alpha_{\boldsymbol{\theta}}$	2×10^{-3}	2×10^{-3}	2×10^{-3}	2×10^{-3}	2×10^{-3}
Horizon $\alpha_{\boldsymbol{\phi}}$	2×10^{-4}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
Critic layers	(64, 64)	(64, 64)	(256, 128)	(256, 128)	(256, 256)
Critic $\alpha_{\boldsymbol{\psi}}$	4×10^{-3}	2×10^{-3}	2×10^{-3}	$5 \times 10^{e-4}$	5×10^{-4}
Critic τ	0.2	0.2	0.2	0.995	0.995

Table 4. Task-specific hyper-parameters. All benchmarked algorithms share the same actor and critic network hyper-parameters with ELU activation functions. AHAC and PPO do not have target critic networks and, as such, do not have τ as a hyper-parameter.

H. Experimental results

In addition to the experimental results in Section 5, here we provide the same results in more detail. Figure 14 depicts step-wise and time-wise reward curves for all experiments. Tables 5 and 6 provide asymptotic (converged) results for all tasks with PPO-normalized and raw rewards, respectively.

	Hopper	Ant	Anymal	Humanoid	SNU Humanoid
PPO	1.00 ± 0.11	1.00 ± 0.12	1.00 ± 0.03	1.00 ± 0.05	1.00 ± 0.09
SAC	0.87 ± 0.16	0.95 ± 0.08	0.98 ± 0.06	1.04 ± 0.04	0.88 ± 0.11
SVG	0.84 ± 0.08	0.83 ± 0.13	0.84 ± 0.19	1.06 ± 0.16	0.75 ± 0.23
SHAC	1.02 ± 0.03	1.16 ± 0.13	1.26 ± 0.04	1.15 ± 0.04	1.44 ± 0.08
AHAC	1.10 ± 0.00	1.41 ± 0.08	1.46 ± 0.06	1.35 ± 0.07	1.64 ± 0.07

Table 5. Tabular results of the asymptotic rewards achieved by each algorithm across all tasks. The results presented are PPO-normalized 50 % IQM and standard deviation across 10 random seeds. All algorithms have been trained until convergence.

	AHAC	SHAC	PPO	SAC	SVG
小轮次		16	5		4
批量大小	8	8	8	32	1024
λ	0.95	0.95	0.95		
γ	0.99	0.99	0.99	0.99	0.99
H - 视界		32	32		3
C - 接触阈值	500				
梯度范数	1.0	1.0	1.0		
ϵ			0.2		
Actor $log(\sigma)$ 边界				(-5,2)	(-5,2)
α - 温度				0.2	0.1
λ_α				10^{-4}	10^{-4}
$\|D\|$- 缓冲区大小				10^6	10^6
种子步数	0	0	0	10^4	10^4

表 *3*。第 5 节中所有基准算法的超参数表。这些参数在所有任务中共享。

	跳跃者	Ant	Anymal	Humanoid	SNU 人形机器人
Actor 层	(128, 64, 32)	(128, 64, 32)	(256, 128)	(256, 128)	(512, 256)
Actor $\alpha_{\boldsymbol{\theta}}$	2×10^{-3}	2×10^{-3}	2×10^{-3}	2×10^{-3}	2×10^{-3}
Horizon $\alpha_{\boldsymbol{\phi}}$	2×10^{-4}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
Critic 层	(64, 64)	(64, 64)	(256, 128)	(256, 128)	(256, 256)
Critic $\alpha_{\boldsymbol{\psi}}$	4×10^{-3}	2×10^{-3}	2×10^{-3}	$5 \times 10^{e-4}$	5×10^{-4}
Critic τ	0.2	0.2	0.2	0.995	0.995

表 *4*。任务特定的超参数。所有基准测试算法共享相同的 Actor 和评判网络超参数，并采用 ELU 激活函数。AHAC 和 PPO 没有目标评判网络，因此不存在作为超参数的 τ。

H. 实验结果

除了第5节中的实验结果外，此处我们提供了更详细的结果。图14 描绘了所有实验的逐步和按时奖励曲线。表 5 和6 分别提供了所有任务在使用 PPO 归一化和原始奖励时的渐近（收敛）结果。

	跳跃者	Ant	Anymal	Humanoid	SNU 人形机器人
PPO	1.00 ± 0.11	1.00 ± 0.12	1.00 ± 0.03	1.00 ± 0.05	1.00 ± 0.09
SAC	0.87 ± 0.16	0.95 ± 0.08	0.98 ± 0.06	1.04 ± 0.04	0.88 ± 0.11
SVG	0.84 ± 0.08	0.83 ± 0.13	0.84 ± 0.19	1.06 ± 0.16	0.75 ± 0.23
SHAC	1.02 ± 0.03	1.16 ± 0.13	1.26 ± 0.04	1.15 ± 0.04	1.44 ± 0.08
AHAC	1.10 ± 0.00	1.41 ± 0.08	1.46 ± 0.06	1.35 ± 0.07	1.64 ± 0.07

表 *5*。各算法在所有任务中实现的渐近奖励的表格结果。所示结果为 PPO 归一化 50% IQM 以及 10 个随机种子的标准差。所有算法均已训练至收敛。

	Hopper	Ant	Anymal	Humanoid	SNU Humanoid
PPO	4742 ± 521	6605 ± 793	12029 ± 360	7293 ± 365	4114 ± 370
SAC	4126 ± 759	6275 ± 528	11788 ± 722	7285 ± 292	3620 ± 453
SVG	3983 ± 379	5482 ± 859	10104 ± 2286	7731 ± 1167	3086 ± 946
SHAC	4837 ± 142	7662 ± 859	15157 ± 481	8387 ± 292	5924 ± 329
AHAC	5216 ± 21	9313 ± 528	17562 ± 722	9846 ± 511	6746 ± 288

Table 6. Tabular results of the asymptotic (end of training) rewards achieved by each algorithm across all tasks. The results presented are 50 % IQM and standard deviation across 10 random seeds. All algorithms have been trained until convergence.

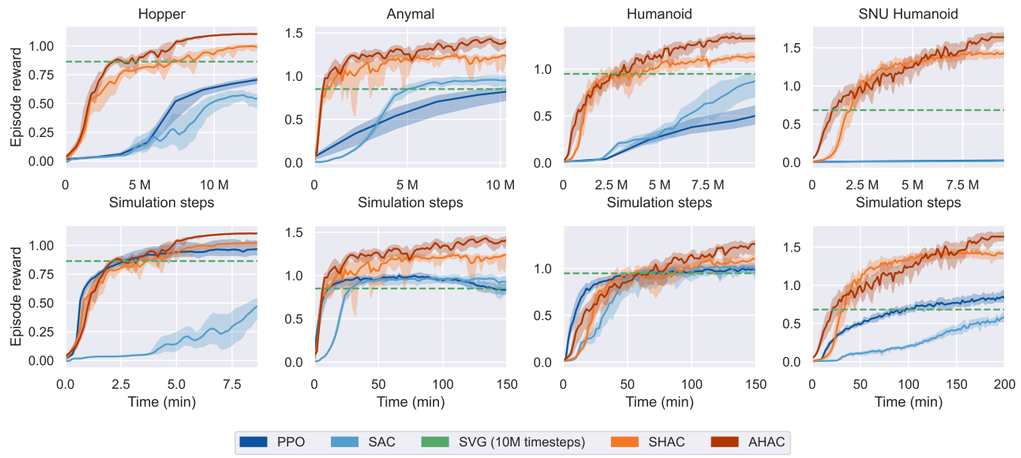


Figure 14. **Reward curves for all tasks against both simulation steps and training time**. We plot 50 % IQM and 95 % CI.

I. Ablation study details

In Section 5, we provided an ablation study of the individual contributions of our proposed approach, AHAC, as summarized in Appendix C. In this section, we provide further details on the conducted experiments. The aim of the study is to understand what changes contribute to the asymptotic performance of AHAC. To best achieve that, we started with SHAC as the baseline, using the tuned version detailed in Appendix G above. Afterwards, we add the individual components that contribute to AHAC using the hyper-parameter from the section above. Note that only hyper-parameters particular to AHAC have been tuned to achieve the results presented in this paper; all other hyper-parameters are the ones tuned to our baseline SHAC with $H = 32$. In particular, we have only tuned the adaptive horizon learning rate α_ψ and contact threshold C. Table 7 shows the detailed differences between the ablations presented in Section 5. The ablations include:

1. SHAC H=32 - our baseline with most hyper-parameters tuned to it.
2. SHAC H=29 - SHAC using the horizon H which AHAC converges to asymptotically.
3. Adaptive Objective - SHAC using the adaptive horizon objective introduced in Eq. 10 but without using it to adapt to the horizon.
4. Adaptive Horizon - SHAC using the objective in Eq 10 and adapting the horizon. This is equivalent to AHAC without the double critic and with iterative training.
5. Iterative critic - SHAC with a single target critic, utilizing iterative critic training until convergence.
6. Double critic - SHAC with a double critic and no target.

Previously in Section 5, we only provided end of training results for the Ant task. In Table 8 we provide the same results in tabular form. We also provide the learning curves for the same experiments in Figure 15.

	跳跃者	Ant	Anymal	Humanoid	SNU 人形机器人
PPO	4742 ± 521	6605 ± 793	12029 ± 360	7293 ± 365	4114 ± 370
SAC	4126 ± 759	6275 ± 528	11788 ± 722	7285 ± 292	3620 ± 453
SVG	3983 ± 379	5482 ± 859	10104 ± 2286	7731 ± 1167	3086 ± 946
SHAC	4837 ± 142	7662 ± 859	15157 ± 481	8387 ± 292	5924 ± 329
AHAC	5216 ± 21	9313 ± 528	17562 ± 722	9846 ± 511	6746 ± 288

表 *6*。各算法在所有任务中实现的渐近（训练结束）奖励的表格结果。所示结果为 10 个随机种子下的 50% IQM 和标准差。所有算法均已训练至收敛。

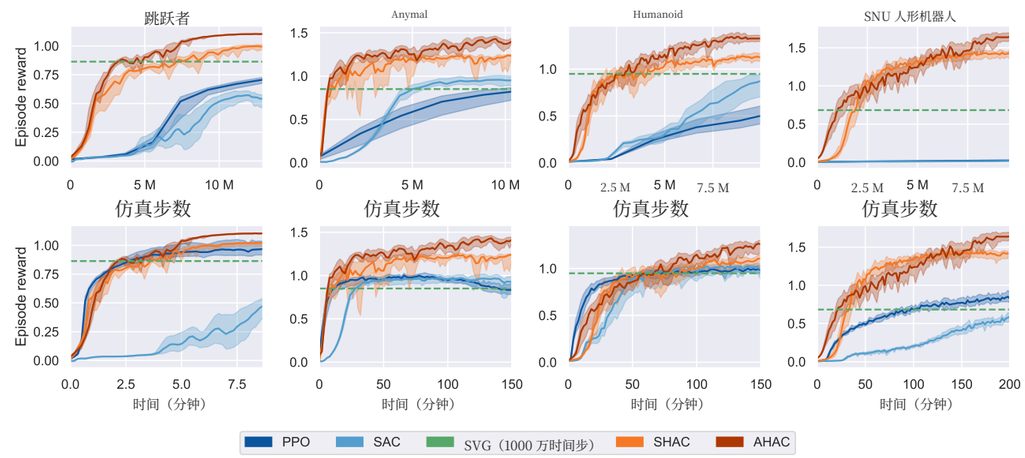


图 *14*。**所有任务针对仿真步数和训练时间的奖励曲线**。我们绘制了 50% IQM 和 95% CI。

I. 消融研究详情

在第 5 节中，我们对我们提出的方法 AHAC 的各项贡献进行了消融研究，总结见附录 C。在本节中，我们将提供所开展实验的更多细节。该研究的目的是了解哪些变化有助于提升 AHAC 的渐近性能。为了最好地实现这一目标，我们以 SHAC 作为基线，使用上文附录 G 中详述的调优版本。随后，我们利用上一节中的超参数，逐个添加构成 AHAC 的各个组件。请注意，仅针对 AHAC 特有的超参数进行了调优，以得出本文呈现的结果；所有其他超参数均采用为我们基线 SHAC 调优后的值，并配合 。具体而言，我们仅调优了自适应视界学习率 α_ψ 和接触阈值 C。表 7 展示了第 5 节中所呈现的各消融实验之间的详细差异。这些消融实验包括：

1. SHAC H=32——我们的基线，其大多数超参数已针对该方法进行调优。
2. SHAC H=29 - 使用 AHAC 渐近收敛到的 Horizon H 的 SHAC。
3. 自适应目标 - 使用公式 10 中引入的自适应视界目标，但不利用其自适应调整视界的 SHAC。
4. 自适应视界 - 使用公式 10 中的目标并自适应调整视界。这等同于不含双重评判器且采用迭代训练的 AHAC。
5. 迭代评判器 - 具有单一目标评论家并利用迭代 Critic 训练直至收敛的 SHAC。
6. 双重评判器 - 具有双重评判器且无目标的 SHAC。

此前在第 5节中，我们仅提供了蚂蚁任务训练结束时的结果。在表 8中，我们以表格形式提供了相同的结果。此外，我们在图 15中提供了相同实验的学习曲线。

	Ablation	H	Actor objective	Critic	Iterative critic training
	SHAC H=32	32	Eq. 8	Single w/ target	
Actor ablations	SHAC H=29	29	Eq. 8	Single w/ target	
	Adapt. Objective	32	Eq. 10	Single w/ target	
	Adapt. Horizon	adaptive	Eq. 10	Single w/ target	
Critic ablations	Iterative critic	32	Eq. 8	Single w/ target	✓
	Double critic	32	Eq. 8	Dual	
	AHAC	adaptive	Eq. 10	Dual	✓

Table 7. Differences between ablations studied, split into actor and critic ablations. All ablations only introduce one component to the baseline, SHAC.

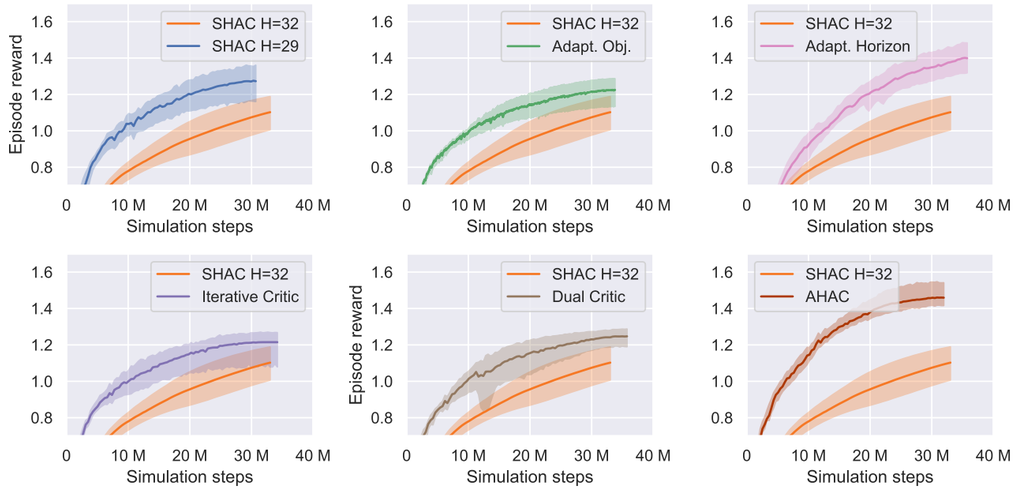


Figure 15. Standalone ablation results for the Ant task. These results are the same as in Figure 11 but presented in a different format for improved legibility.

Ablation	Asymptotic reward
SHAC H=32	1.16 ± 0.14
1. SHAC H=29	1.23 ± 0.17
2. Adaptive Objective	1.18 ± 0.18
3. Adaptive Horizon	1.35 ± 0.12
4. Iterative Critic	1.17 ± 0.13
5. Double Critic	1.20 ± 0.07
AHAC	1.41 ± 0.08

Table 8. Results of asymptotic performance of our ablation study showing 50% IQM and standard deviation.

	消融实验	H	Actor 目标函数	Critic	迭代 Critic 训练
	SHAC H=32	32	公式8	Single w/ target	
Actor 消融实验	SHAC H=29	29	公式 8	Single w/ target	
	自适应目标	32	公式 10	Single w/ target	
	自适应时域	自适应的	公式 10	Single w/ target	
Critic 消融实验	迭代评判器	32	公式 8	Single w/ target	✓
	双重评判器	32	公式 8	Dual	
	AHAC	自适应的	公式 10	Dual ✓	

表 7。所研究的消融实验之间的差异，分为 Actor 消融实验和 Critic 消融实验。所有消融实验仅在基线 SHAC 中引入一个组件。

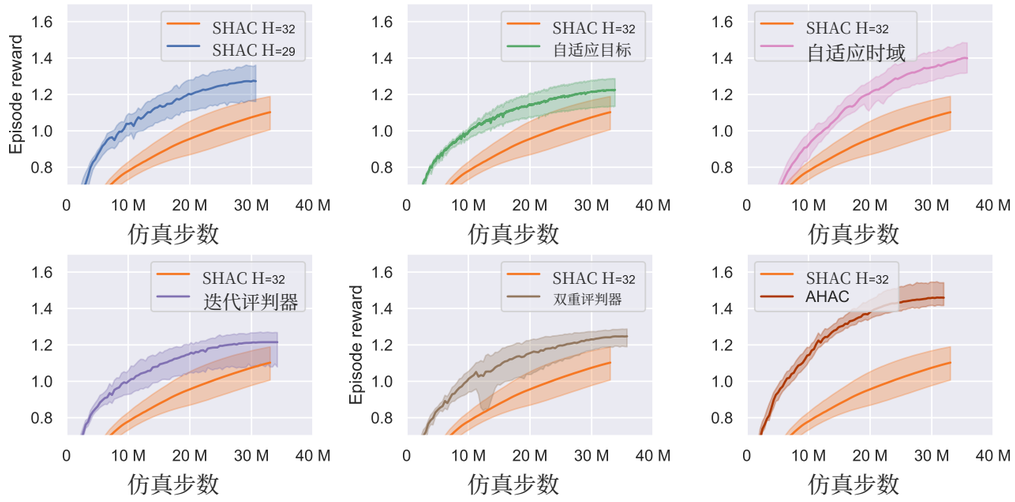


图 *15*。蚂蚁任务的独立消融实验结果。这些结果与图11 中的结果相同，但采用了不同的格式以提高可读性。

消融实验	渐近奖励
SHAC H=32	1.16 ± 0.14
1. SHAC H=29	1.23 ± 0.17
2. 自适应目标	1.18 ± 0.18
3. 自适应视界	1.35 ± 0.12
4. 迭代评判器	1.17 ± 0.13
5. 双重评判器	1.20 ± 0.07
AHAC	1.41 ± 0.08

表 *8*。我们的消融研究的渐近性能结果，展示了 50% IQM 和标准差。